

Contracting structures in public procurement: Evidence from donor-funded electrification in Kenya

On-line Appendix

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Authors are in ©[Certified Random](#) order. Wolfram: Massachusetts Institute of Technology and NBER. Miguel: University of California, Berkeley and NBER. Hsu: University of California, Berkeley. Berkouwer: University of Pennsylvania and NBER. Paper [available here](#).

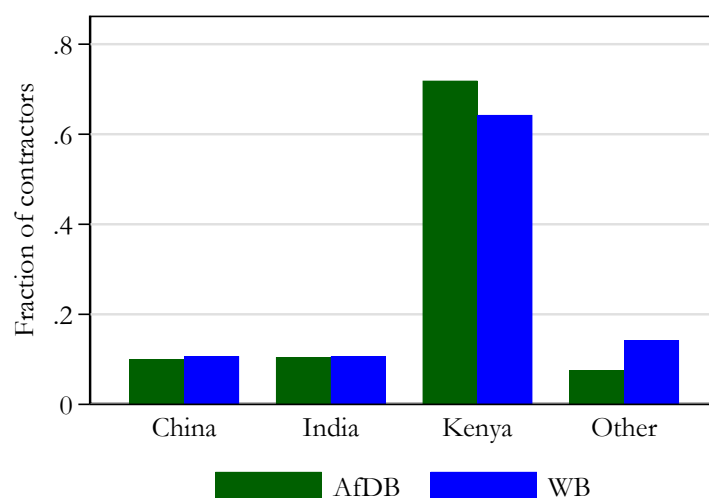
A Appendix Figures

Figure A1: Readyboards installed by WB and AfDB-funded contractors



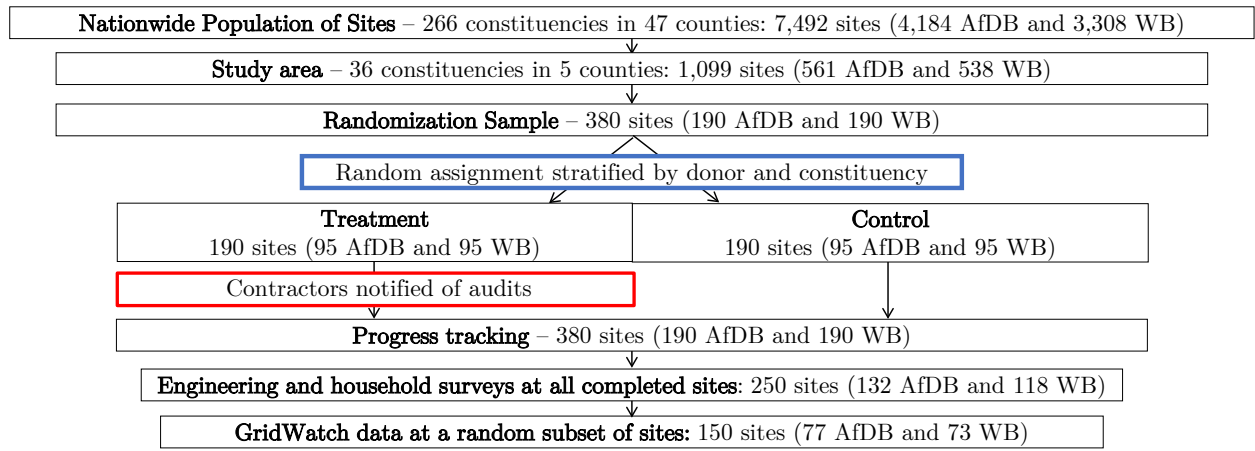
Notes: Panel A displays a ready board provided by a WB-funded contractor. Panel B displays a ready board provided by an AfDB-funded contractor.

Figure A2: World Bank contractors and African Development Bank subcontractors by country of origin



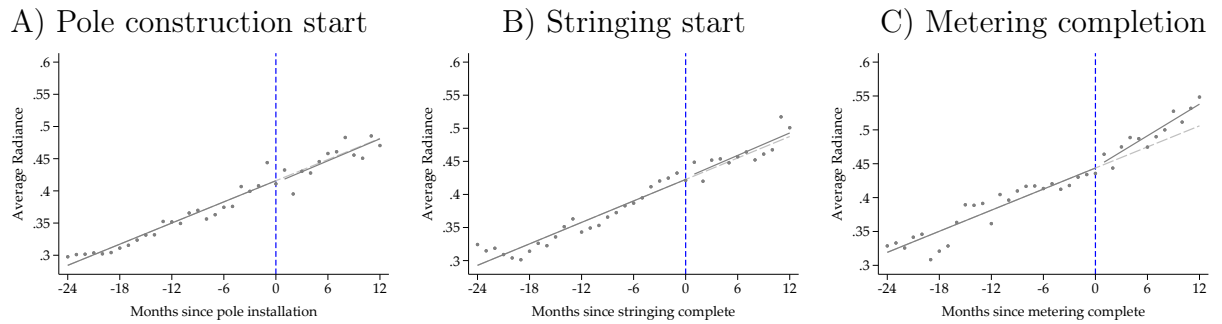
Notes: 59 companies were either awarded World Bank (WB) contracts or were approved to subcontract with one of the African Development Bank (AfDB) contractors for the procurement of poles, conductors, cables, or installation. This graph shows the distribution of countries of origin of these 59 companies. AfDB subcontractors are inverse-weighted by the number of good-specific subcontractors for which that AfDB contractor got approval, as most likely only one was used per good.

Figure A3: Project design



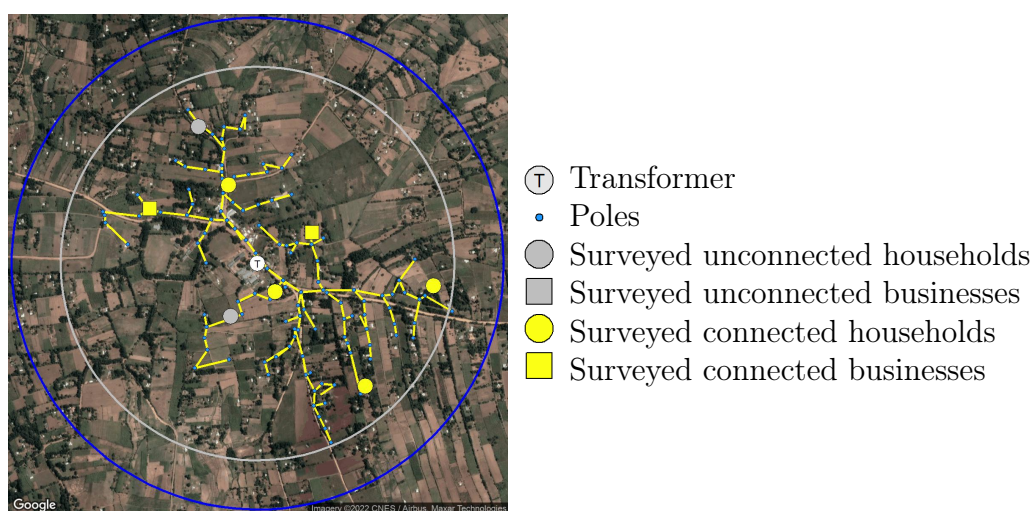
Notes: Sample selection and randomization, starting with the nationwide sample of sites selected for the Last Mile Connectivity Project by funding source (African Development Bank or World Bank). We randomly select 380 out of the 1,099 sites located in the five study counties, randomly assign each to control or treatment, and complete engineering assessments and household surveys at all 250 sites where there had been meaningful construction by the end of surveying activities in mid 2021. Construction progress was tracked at the remaining sites through mid 2022.

Figure A4: Event study: nightlights after construction progress



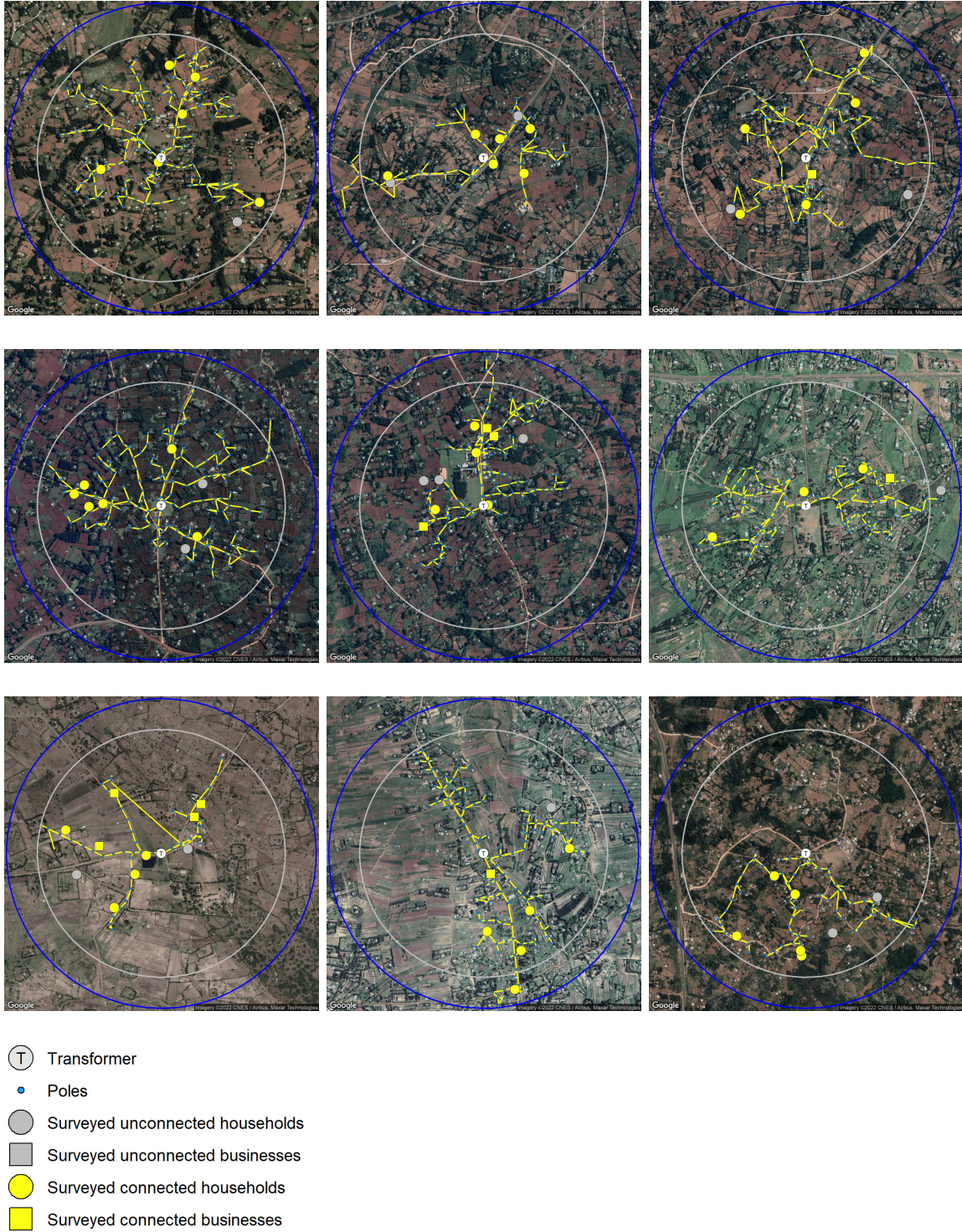
Notes: Data on construction progress collected through phone surveys with local village representatives. As expected, nighttime radiance data (Elvidge et al., 2017) increases after metering completion (when the electricity connection is activated) but not earlier.

Figure A5: Infrastructure data collected (example site)



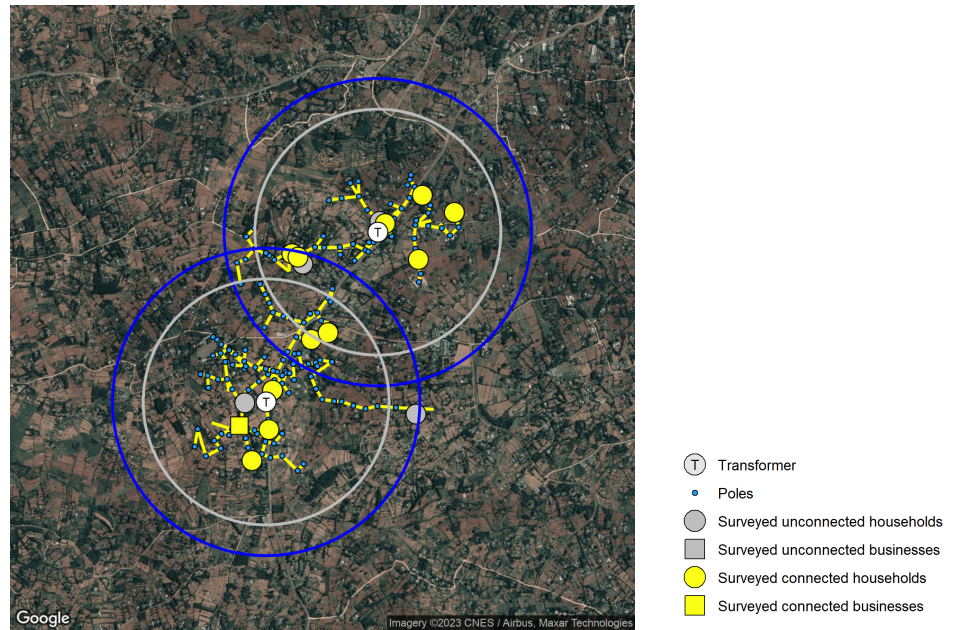
Notes: Construction data collected at an example site ([Figure A6](#) shows additional examples). The grey (blue) line denotes 600 (700) meters from the transformer. The engineering surveys record the locations of poles (blue dots), conductors (yellow lines), and infrastructure quality. At each site, between 4 to 9 connected and unconnected residential compounds and firms were randomly selected to participate in the socioeconomic survey and to receive Grid-Watch devices to measure power quality: these are marked with yellow and gray circles and squares. (Note that random spatial noise has been added to preserve respondent anonymity here.)

Figure A6: Engineering data collected (additional example sites)



Notes: These maps display the construction data collected at example sites. The grey line denotes 600 meters and the blue line denotes 700 meters from the transformer ('T') at the center. [Section 4](#) provides additional information on data collection. To preserve anonymity, random spatial noise has been added to household and business locations.

Figure A7: Two sites located less than 1,200 meters apart



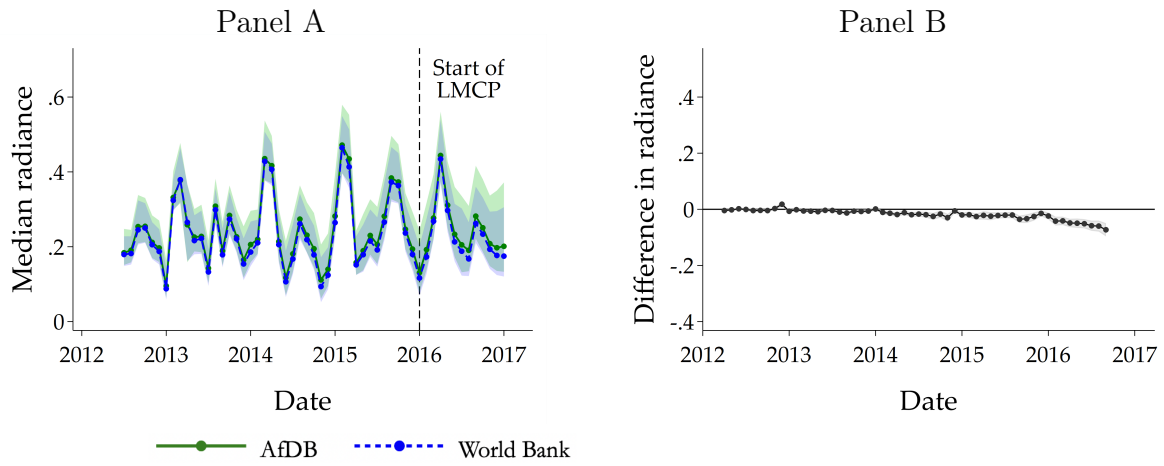
Notes: This map displays two sites whose transformers are located 990 meters apart, such that the 600 and 700 meter radius eligibility areas overlap. See [Section 4](#) for a discussion on this issue. To preserve anonymity, random spatial noise has been added to household and business locations.

Figure A8: A GridWatch device



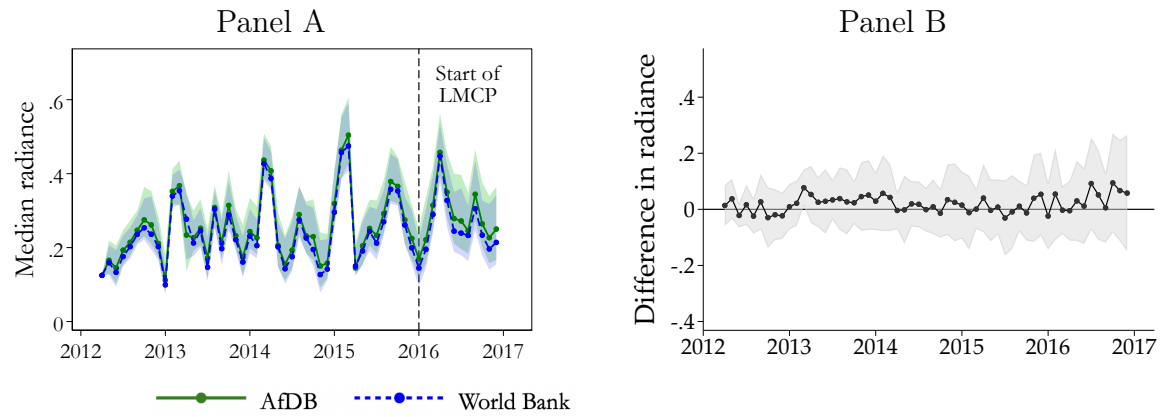
Notes: A GridWatch device, part of nLine's technologies used to measure household-level power outages and voltage. The device transmits data to the cloud in near real-time over the cellular network, and stores data locally to transmit later in the case of network failure. The GridWatch server consolidates data to detect patterns in power outages and reduce noisy signals.

Figure A9: Site-level nighttime radiance by funding source, national



Notes: Panel A presents median monthly nighttime radiance from the Visible Infrared Imaging Radiometer Suite (VIIRS) between 2012-2017 per month, with bands showing the 25th to 75th percentile across sites, before and after the start of the Last Mile Connectivity Project (LMCP). Panel B shows imbalance that is statistically significant in later years, but economically small across World Bank and African Development Bank-funded sites (estimates include constituency fixed effects). [Table B3](#) demonstrates baseline balance using a pooled regression of these data. [Figure A10](#) performs the same analysis on the study sample of transformers and finds no statistically detectable relationship between radiance and funder assignment.

Figure A10: Site-level nighttime radiance by funding source, study sites



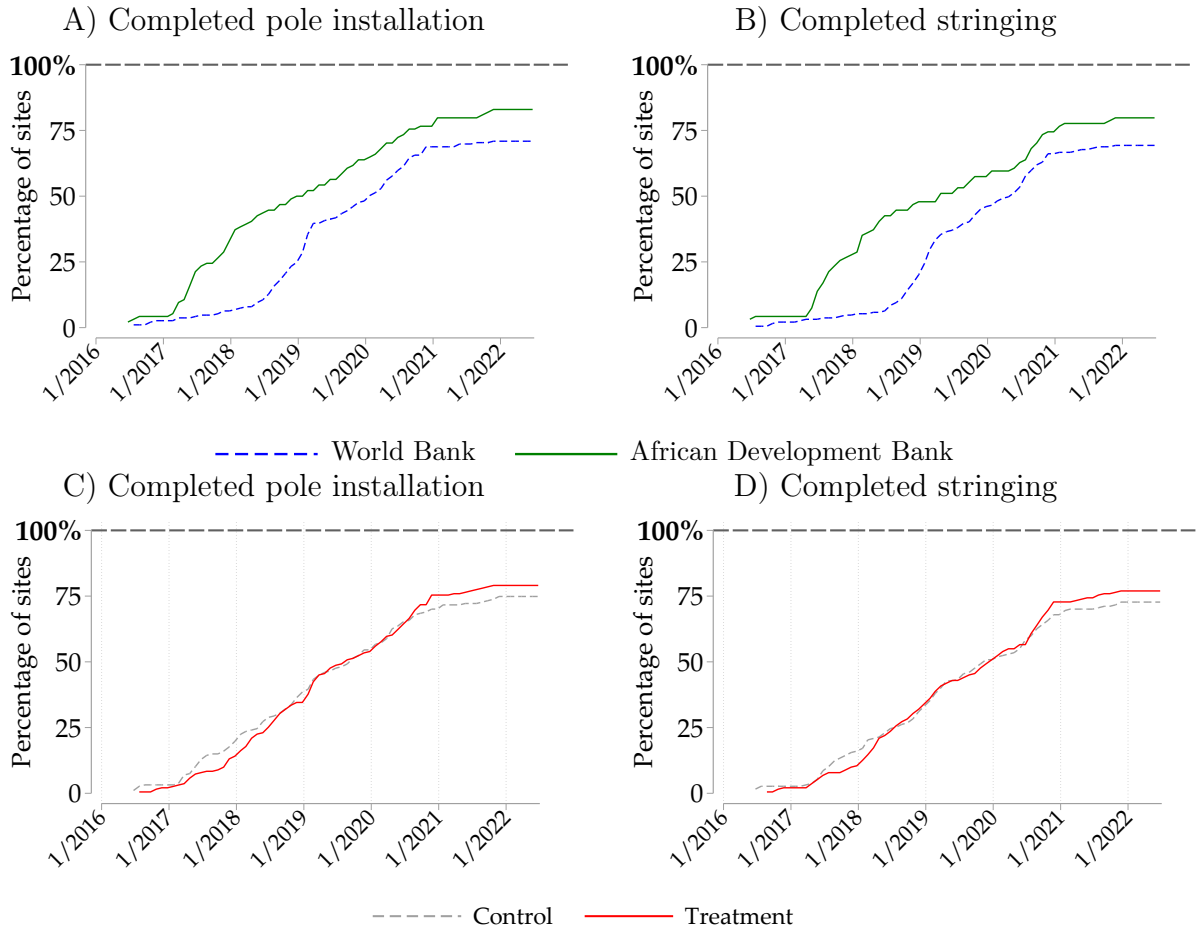
Notes: Panel A presents median monthly nighttime radiance (VIIRS), with bands showing the 25th to 75th percentile across sites, before and after the start of the LMCP. Panel B confirms that radiance is statistically indistinguishable across sites (estimates include constituency FE). [Table D1](#) confirms baseline balance using a pooled regression of these data. [Figure A9](#) performs the same analysis on the national sample of phase 1 transformers.

Figure A11: Monitoring Intervention

 THE WORLD BANK IBRD • IDA	 Kenya Power	
Contractor XYZ ADDRESS P.O. Box YYY-ZZZ Nairobi, Kenya		June 2017
TO: CONTRACTOR NAME RE: ENHANCED MONITORING PROGRAM ("EMP") FOR LMCP MAXIMIZATION SITES		
Dear Sir/Madame: Kenya Power aims to provide the highest quality of electricity to all Kenyans. To achieve this goal, an international team of engineers will closely audit the quality of construction at a number of Last Mile Connectivity Project ("LMCP") maximization sites. These independent audits will be performed as part of the Enhanced Monitoring Program ("EMP"), and will target both African Development Bank and World Bank project sites. The results of the EMP audits will be shared with project supervisors, financiers, and international agencies, all of which may impose consequences on future contracting opportunities, as they see fit. Upon project completion, EMP technicians will extensively measure the quality of various aspects of construction, including: - Distance between poles - Line sag - Quality of connection between transformer and LV wiring - Blackouts and electricity reliability post-connection We wish to inform you of the sites that have been awarded to you that have been selected for the EMP. Please find attached a list of these sites. Sincerely yours,		
 _____ I [redacted] Senior Energy Specialist The World Bank	 _____ S [redacted] Principal Power Engineer The African Development Bank	 _____ J [redacted] Electrification Project Manager Kenya Power & Lighting Company

Notes: This figure displays the monitoring intervention sent to contractors. All letters were signed by relevant representatives from Kenya Power, the World Bank, and the African Development Bank, with their names and positions listed below. Each letter specified the contractor's name and contact information. The letters were then hand-delivered to management at the relevant contractors by members of our research team to ensure receipt, together with the list of treatment sites referenced in the letter.

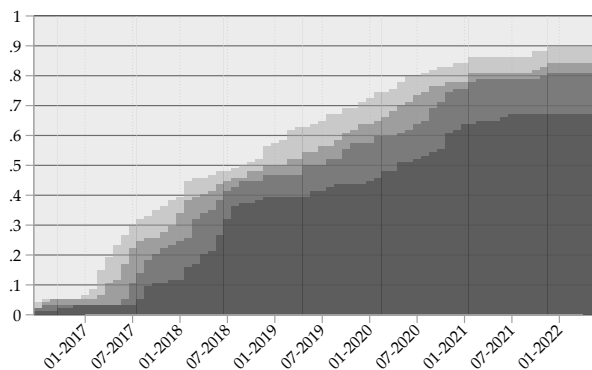
Figure A12: Construction progress by funding source



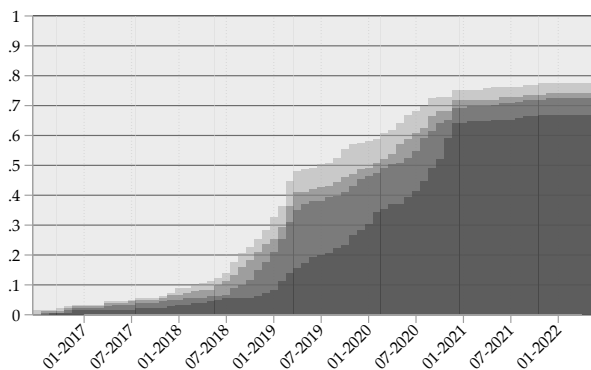
Notes: Construction progress data for sites located in the five study counties collected through phone surveys with village representatives. Panels A and B separate progress by whether the site was constructed under bundled contracting without additional monitoring (observed at African Development Bank sites) and under unbundled contracting with additional monitoring (observed at World Bank sites). Panels C and D separate progress by audit treatment status. [Figure 4](#) displays progress for started construction and completed metering.

Figure A13: Construction progress by contracting and monitoring status

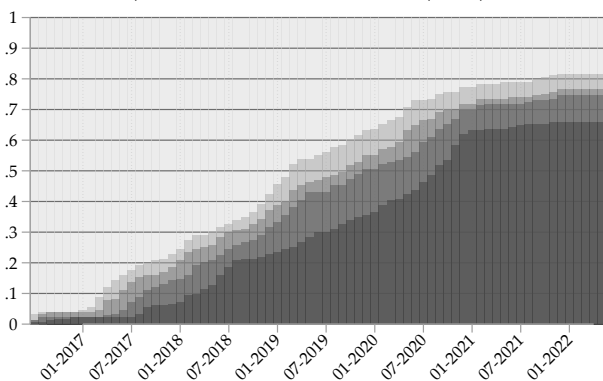
A) Sites with no unbundling or monitoring
(188)



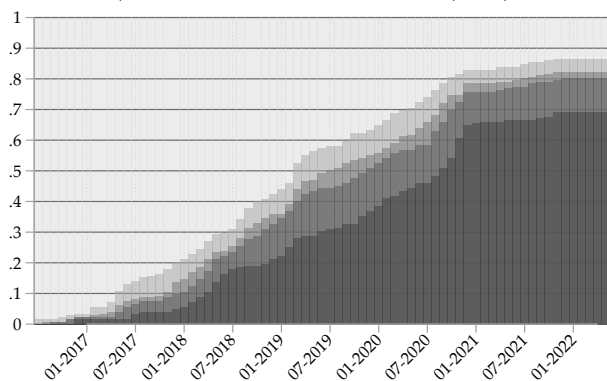
B) Sites with unbundling and monitoring
(181)



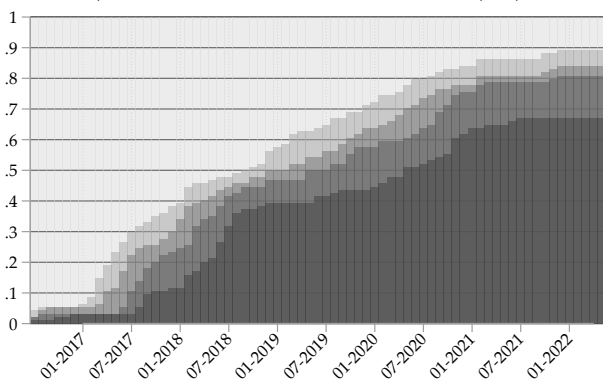
C) Audit control sites (184)



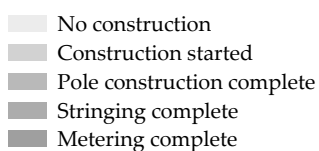
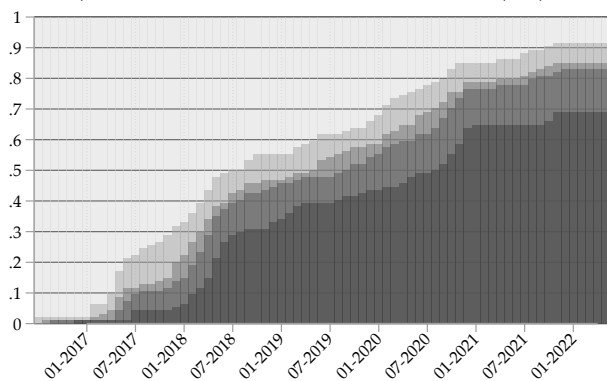
D) Audit treatment sites (185)



E) Audit control AfDB sites (94)

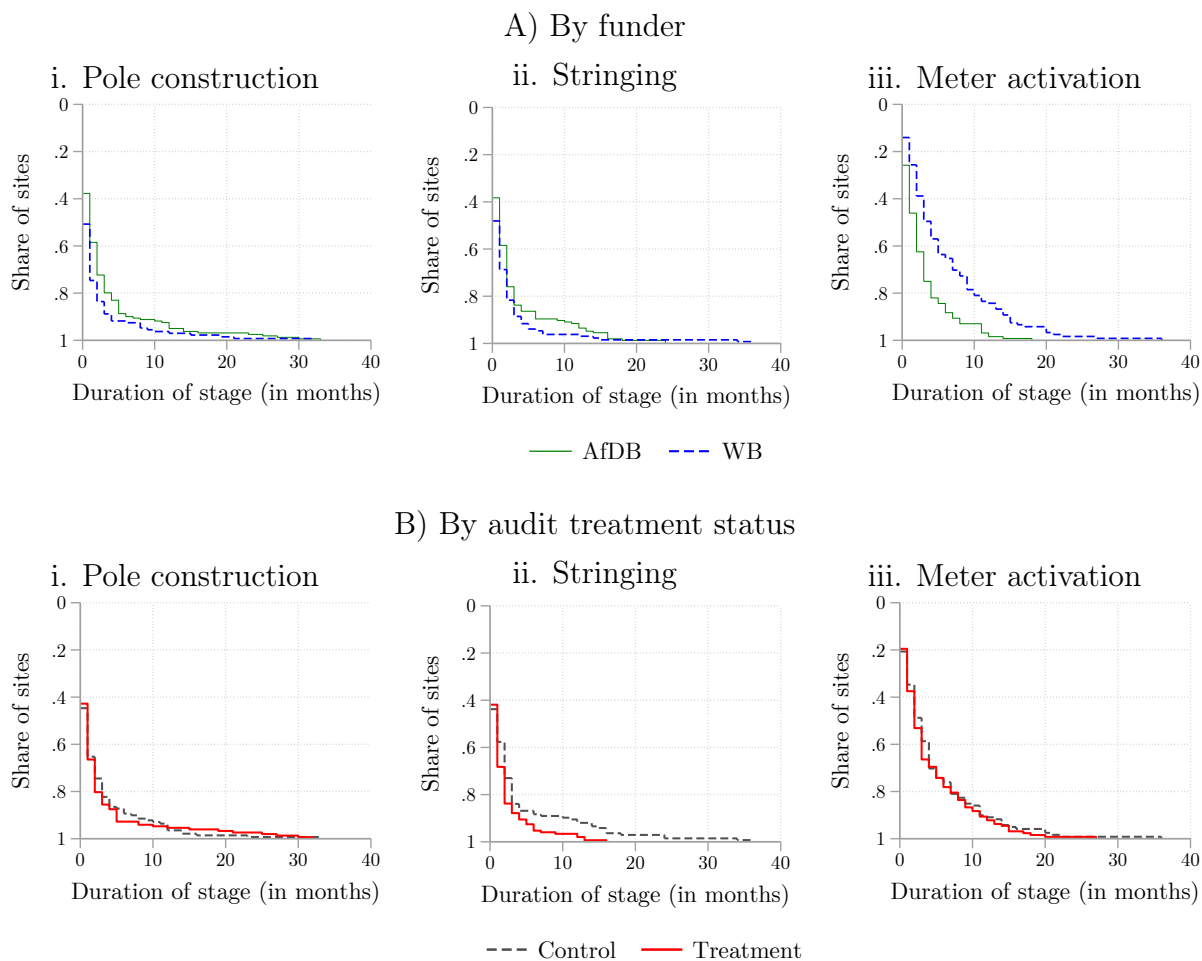


F) Audit treatment AfDB sites (94)



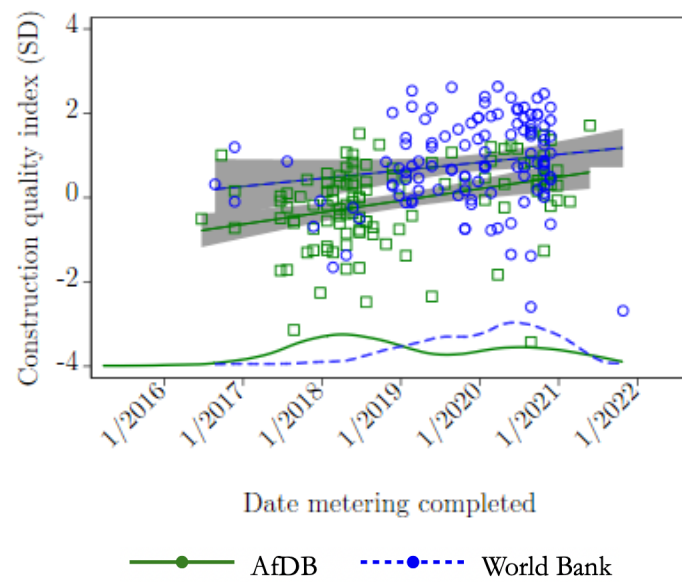
Notes: Construction progress data for the four stages of construction, collected through recurring phone surveys with village representatives, separately by unbundling, monitoring, and audit treatments.

Figure A14: Conditional construction progress by funder and by treatment status



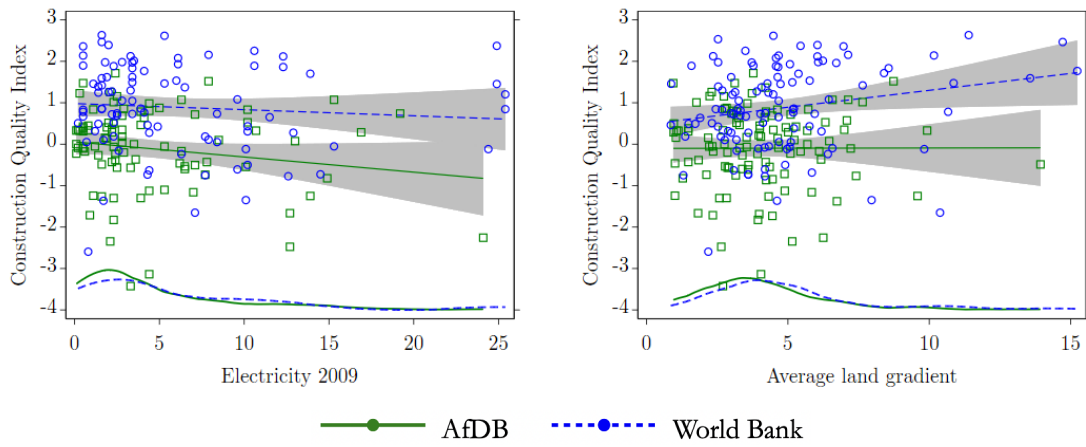
Notes: Duration of each construction stage, conditional on having completed the preceding stage, separately by funder (Panel A) and audit treatment status (Panel B).

Figure A15: Delays and construction quality (Outcome 1)



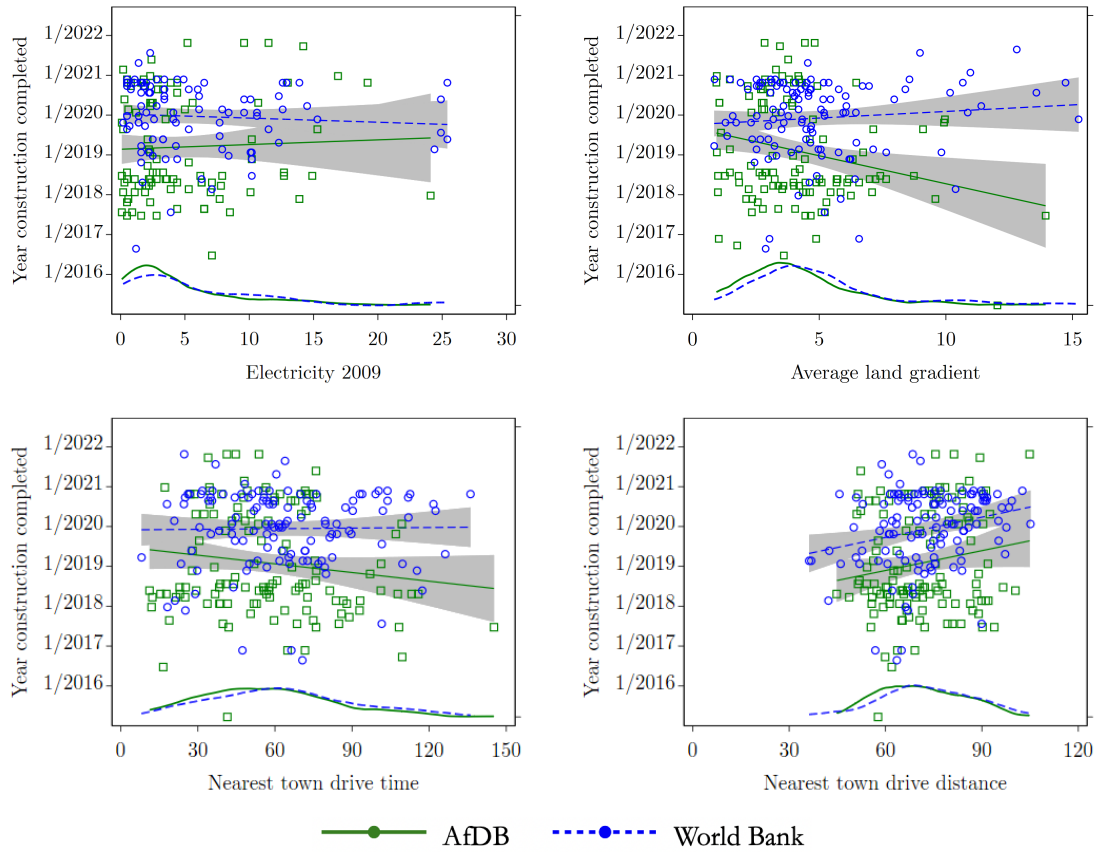
Notes: Difference in construction quality by funder and by when metering was completed at a site.

Figure A16: Construction Quality, Gradient, and Electricity Access



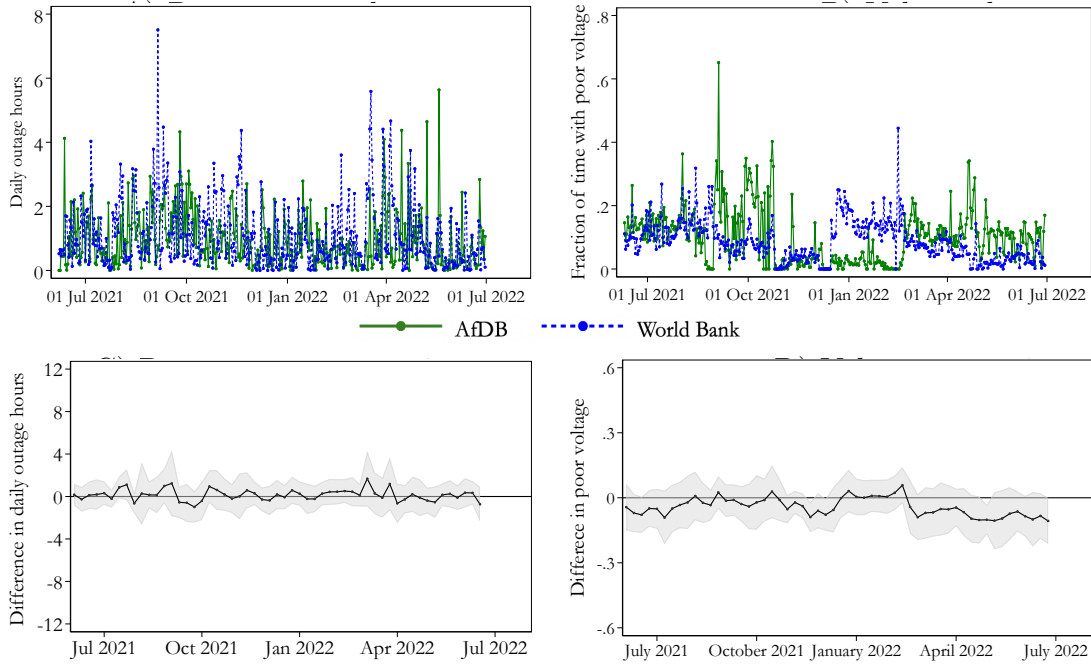
Notes: Differences in construction quality between WB and AfDB are approximately constant across the distribution of baseline electricity access and land gradient. Baseline electricity data from Kenya National Bureau of Statistics (2006; 2009). Average land gradient is calculated for each site over the 600 meter radius around its transformer using the 90-meter Shuttle Radar Topography Mission Global Digital Elevation Model.

Figure A17: Construction Delays and Geographic Features



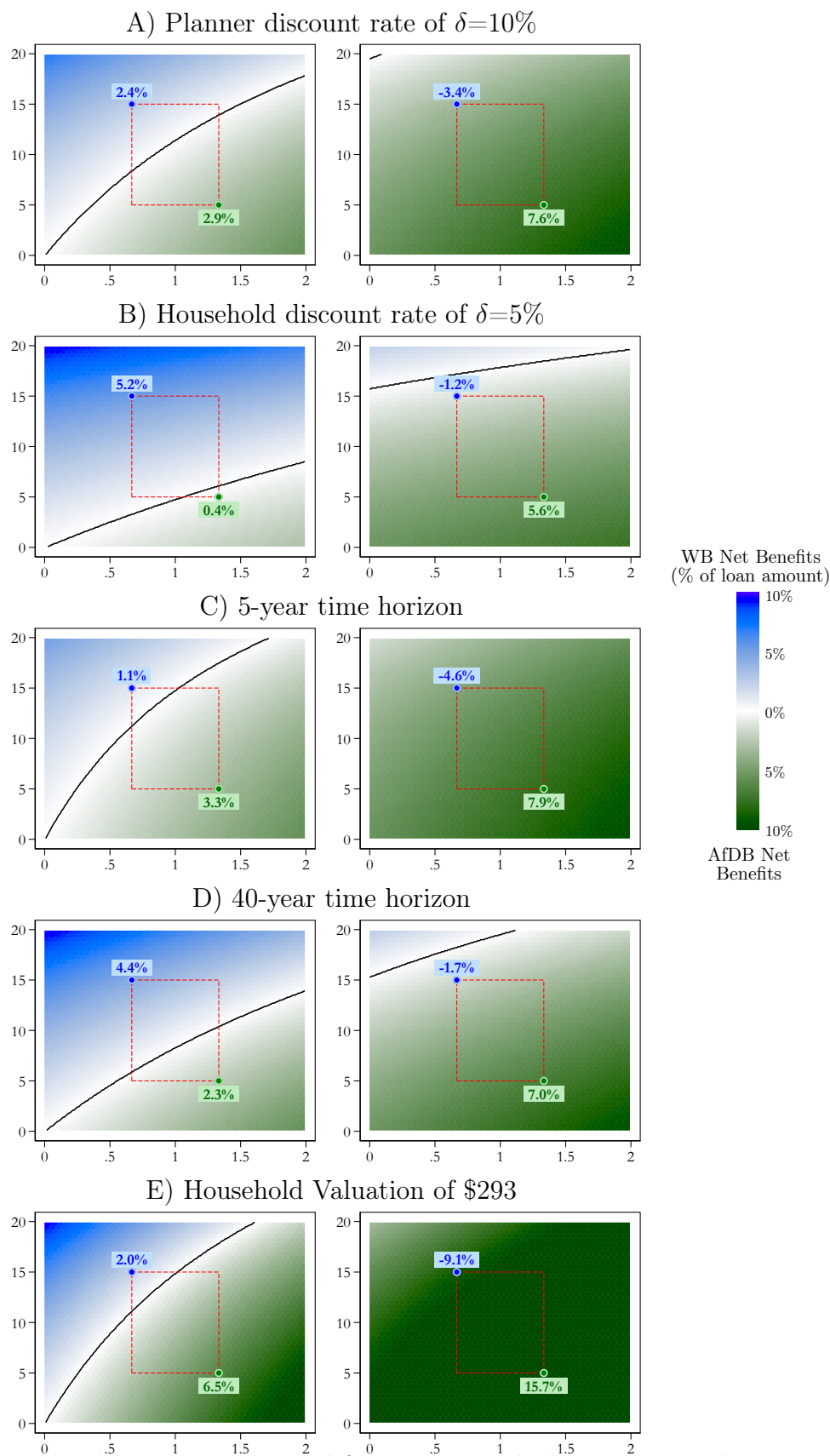
Notes: Differences in construction delays between WB and AfDB are approximately constant across the distribution of baseline electricity access, land gradient, and distance to nearest town. Baseline electricity data from KNBS (2006; 2009). Land gradient from USGS (2018). Towns from WRI (2007); distances calculated using HERE (2022).

Figure A18: Reliability and voltage quality by funding source



Notes: Panels A and B present the hours of power outage per day and fraction of time experiencing poor voltage quality, respectively, for World Bank and African Development Bank sites. Panels C and D estimate a separate coefficient for each week of the sample, with constituency fixed effects and standard errors clustered by site. In the voltage graphs, periods with power outages are set to missing in the voltage measurement data, but the results look similar when coding such periods as having $V = 0$.

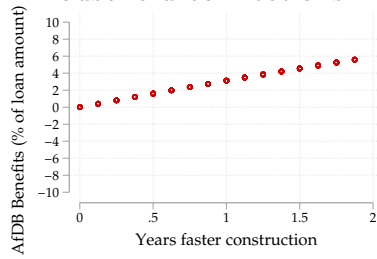
Figure A19: Costs versus benefits on various assumptions



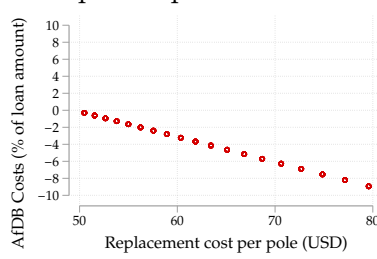
Notes: Variations on the assumptions used in Figure 5, which presents results using our preferred assumptions. Each sub-title indicates the one aspect that has been changed relative to Figure 5.

Figure A20: Components of cost-benefit analyses

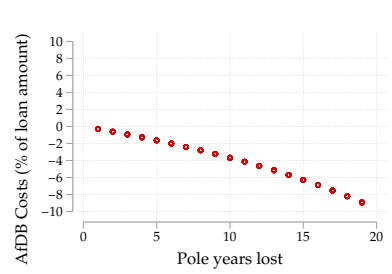
i. AfDB benefit from earlier household connections



ii. AfDB cost from increased pole replacement



iii. AfDB cost from pole years lost



B Appendix Tables

Table B1: Representative examples of procurement language

(1) World Bank	(2) African Development Bank	(3)
<i>Development and maintenance of a procurement plan</i>		
[1.18 ^a] [...] As part of the project preparation, the Borrower shall prepare a preliminary procurement plan, however tentative, for the entire scope of the project. At a minimum, the Borrower shall prepare a detailed and comprehensive procurement plan including all contracts for which procurement action is to take place in the first 18 (eighteen) months of project implementation. An agreement with the Bank shall be reached at the latest during loan negotiations. The Borrower shall update procurement plans throughout the duration of the project at least annually by including contracts previously awarded and to be procured in the next 12 (twelve) months. All procurement plans and their updates or modifications shall be subject to the Bank's prior review and no objection before implementation. [...]	[1.16 ^b] As part of the preparation of the project the Borrower shall prepare and, before negotiating the Financing Agreement, furnish to the Bank for its approval, a Procurement Plan acceptable to the Bank setting forth: (a) the particular contracts for the goods, works, and/or services required to carry out the project during the initial period of at least 18 months; (b) the proposed methods for procurement of such contracts that are permitted under the Financing Agreement, and (c) the related Bank review procedures. The Borrower shall update the Procurement Plan annually or as needed throughout the duration of the project. The Borrower shall implement the Procurement Plan in the manner in which it has been granted the approval by the Bank.	0.85
<i>Preference for domestic suppliers</i>		
[2.55 ^a] At the request of the Borrower, and as stipulated in the agreed Procurement Plan and set forth in the bidding documents, a margin of preference may be provided in the evaluation of bids for: (a) goods manufactured in the country of the Borrower when comparing bids offering such goods with those offering goods manufactured abroad; and (b) works in member countries below a specified threshold of GNP per capita, when comparing bids from eligible domestic contractors with those from foreign firms.	[2.55 ^b] At the request of the Borrower, and under conditions to be agreed under the Financing Agreement and set forth in the bidding documents, a margin of preference may be provided in the evaluation of bids for: (a) goods manufactured either in the country of the Borrower (domestic), or in a country which has joined the borrowing country in a regional economic institutional arrangement (regional), when comparing bids offering such goods with those offering goods manufactured abroad; and (b) contractors either from the country of the Borrower (domestic), or from Member Countries which have joined the borrowing country in a regional economic institutional arrangement (regional), when comparing bids from eligible domestic/regional contractors with those from foreign firms.	0.89
<i>Evidence of financial sustainability</i>		
[3.1.a.i ^c] Historical financial performance; by Submission of audited balance sheets or if not required by the law of the bidder's country, other financial statements acceptable to the Employer, for the last five (5) years to demonstrate the current soundness of the bidders financial position and its prospective long term profitability.	[2.3.1 ^d] Historical financial performance; Submission of audited balance sheets or if not required by the law of the bidder's country, other financial statements acceptable to the Employer, for the last five [5] years to demonstrate the current soundness of the bidders financial position and its prospective long term profitability.	0.99

Table B2: Summary statistics

	Mean	SD	25 th	50 th	75 th	N
Transformer missing fuse	0.23	0.42	0	0	0	250
Number of transformer lines	3.13	0.99	3	3	4	250
Number of poles	84.92	35.16	58	80	106	250
Number of leaning poles (<85deg)	1.69	2.57	0	1	2	250
Number of cracked poles	20.29	18.01	6	15	29	250
Number of poles without a cap	40.17	28.80	19	34	56	250
Number of stays	54.91	24.34	37	52	70	250
Households surveyed	3.78	1.63	3	4	5	250
Connected households surveyed	3.15	1.64	2	3	4	250
Year households connected	2018.89	1.13	2018	2019	2020	184

Notes: Summary statistics for surveyed sites. The question on connection year was added to the survey later, after surveying had already been completed at 66 sites.

Table B3: Geographic balance of World Bank- and African Development Bank-funded sites

	Road Distance		VIIRS Radiance	Land Gradient
	(1)	(2)	(3)	(4)
World Bank (=1)	1.58** (0.74)	0.63 (0.45)	-0.01 (0.02)	-0.00 (0.07)
Observations	6130	6130	337205	6131
Month FE	No	No	Yes	No
Constituency FE	Yes	Yes	Yes	Yes
AfDB Mean	54.07	34.58	.58	4.24
Outcome variable	Minutes	KM		

Notes: Balance across all sites ([Table D1](#) shows only study sites). All regressions include constituency FE. Columns (1) and (2) estimate road distance from each site to the nearest ‘major town’ (WRI, [2007](#); HERE, [2022](#)). Column (3) estimates monthly average site-level nighttime radiance from July 2012 through January 2017 measured using VIIRS averaged across the 600 meter radius with SE clustered by site (Elvidge et al., [2017](#)). [Figure A10](#) shows the time series at at study sites ([Figure A9](#) shows the same nationwide). Column (4) estimates average site-level land gradient recorded using the 90-meter Shuttle Radar Topography Mission Global Digital Elevation Model. * ≤ 0.10 , ** $\leq .05$, *** $\leq .01$.

Table B4: Transformer facility type

<i>Panel A) Sample field data</i>			
	N	AfDB Mean (SD)	WB (SE)
Health center	250	0.05 (0.22)	-0.00 (0.03)
School	250	0.50 (0.50)	-0.13* (0.07)
Market center	250	0.17 (0.38)	0.09* (0.05)
Religious building	250	0.20 (0.40)	-0.10* (0.05)
Other	250	0.08 (0.28)	-0.03 (0.04)
None	250	0.27 (0.44)	0.12* (0.06)
<i>Panel B) Sample administrative data</i>			
	N	AfDB Mean (SD)	WB (SE)
Health center	378	0.06 (0.24)	-0.03 (0.02)
School	378	0.09 (0.29)	0.18*** (0.04)
Market center	378	0.13 (0.33)	0.03 (0.04)
Religious building	378	0.05 (0.22)	-0.03 (0.02)
Other	378	0.09 (0.29)	0.03 (0.03)
None	378	0.08 (0.27)	0.29*** (0.04)
<i>Panel C) Nationwide administrative data</i>			
	N	AfDB Mean (SD)	WB (SE)
Health center	7396	0.03 (0.18)	-0.02*** (0.00)
School	7396	0.05 (0.23)	-0.01** (0.01)
Market center	7396	0.16 (0.37)	0.01 (0.01)
Religious building	7396	0.02 (0.13)	0.00 (0.00)
Other	7396	0.38 (0.49)	0.22*** (0.01)
None	7396	0.00 (0.00)	0.00 (.)

Notes: Most transformers were constructed between 2005-2015 as part of a push by Kenya's Rural Electrification Authority (REA) to electrify public facilities like schools, religious buildings and markets, and there appear to be some differences in the likelihood of transformers located near specific types of facilities to be assigned to one funder or the other. We test whether transformers connected to certain types of facilities were more or less likely to be assigned to WB or AfDB funding. Total shares can exceed 1 because some transformers

Table B5: Balance in 2009 census socioeconomic characteristics by number of LMCP sites per ward (nationwide sample)

	Dep. Var. Mean [SD]	Percent of LMCP Sites that are WB-funded	N
<i>Panel A: 2009 Census</i>			
Age 14 or Under	49.56 [7.23]	0.01* (0.00)	1094
Consumption	3010.52 [1365.28]	-1.59 (1.25)	1103
Primary Education	57.14 [9.93]	0.00 (0.01)	1110
Secondary Education	20.29 [10.02]	-0.02** (0.01)	1110
Solar Home System	2.00 [2.18]	-0.00 (0.00)	1110
Electricity	11.29 [16.93]	-0.02 (0.01)	1110
High-Quality Wall	16.75 [21.04]	0.01 (0.01)	1110
High-Quality Roof	74.79 [23.44]	-0.03* (0.01)	1110
Population	23527.17 [7765.82]	-8.43 (8.41)	1110
Land Area (sq km)	280.11 [784.28]	0.27 (0.80)	1117
<i>Panel B: Political and Ethnic Divisions</i>			
2013 Kenyatta Share (%)	47.77 [40.58]	0.02 (0.01)	1006
Voted pro-MP in 2013	0.70 [0.46]	-0.00* (0.00)	806
Luo-aligned	0.15 [0.36]	0.00 (0.00)	1108
Kikuyu-aligned	0.28 [0.45]	0.00* (0.00)	1108
Kalenjin-aligned	0.16 [0.36]	0.00 (0.00)	1108
Joint F-test		p-value = .02	

Notes: This table tests for correlations between the share of Last Mile Connectivity Project (LMCP) sites in a ward allocated to World Bank (WB) funding and baseline characteristics, at the ward level, among wards with at least 1 LMCP site. [Table B7](#) shows a version testing balance among study sites only. Panel A tests balance on characteristics from the 2009 national census. Row 1 shows population share aged 14 years or younger. Row 2 shows monthly consumption expenditures per capita in Kenya Shillings (Ksh). Rows 3 and 4 show primary and secondary school completion rates, respectively. Rows 5–8 show percentage of households with solar, electricity, a high wall and roof quality, respectively. Rows 8–9 show the total ward population and land area, respectively. Panel B tests balance on political and ethnic data. Row 10 show the percentage of the ward that voted for Kenyatta in the 2013 presidential election. Row 11 is a dummy variable equal to one if the ward voted for the same party as their constituency MP. Rows 12–14 are dummy variables showing political alignment with major ethnic groups, equal to one if the ward voted for a member of that ethnic group in the 2013 presidential election. All socioeconomic characteristics are for the 2009 census.

Table B6: Balance in 2009 census socioeconomic characteristics by number of LMCP sites per ward – standardized to mean zero, standard deviation one (nationwide sample)

	Dep. Var. Mean [SD]	Proportion of LMCP Sites that are WB-funded	N
<i>Panel A: 2009 Census</i>			
Age 14 or Under	-0.00 [1.00]	0.10* (0.06)	1094
Consumption	0.00 [1.00]	-0.12 (0.09)	1103
Primary Education	0.00 [1.00]	0.04 (0.06)	1110
Secondary Education	-0.00 [1.00]	-0.16** (0.07)	1110
Solar Home System	-0.00 [1.00]	-0.00 (0.07)	1110
Electricity	-0.00 [1.00]	-0.09 (0.08)	1110
High-Quality Wall	-0.00 [1.00]	0.04 (0.06)	1110
High-Quality Roof	-0.00 [1.00]	-0.11* (0.06)	1110
Population	0.00 [1.00]	-0.11 (0.11)	1110
Land Area (sq km)	0.00 [1.00]	0.03 (0.10)	1117
<i>Panel B: Political and Ethnic Divisions</i>			
2013 Kenyatta Share (%)	0.00 [1.00]	0.05 (0.03)	1006
Voted pro-MP in 2013	-0.00 [1.00]	-0.25* (0.14)	806
Luo-aligned	0.00 [1.00]	0.03 (0.07)	1108
Kikuyu-aligned	-0.00 [1.00]	0.14* (0.08)	1108
Kalenjin-aligned	0.00 [1.00]	0.07 (0.08)	1108

Notes: This table replicates [Table B5](#) but standardizes the outcome variables to have a mean of 0 and a standard deviation of 1, and the independent variable from 0 to 1 (in terms of the fraction of sites that are funded by the WB). Thus, each coefficient can be interpreted as the standard deviation change if the ward went from having 0% of its LMCP sites funded by the WB to having 100% of its LMCP sites funded by the WB. * ≤ 0.10 , ** $\leq .05$, *** $\leq .01$.

Table B7: Balance in 2009 census socioeconomic characteristics by number of LMCP sites per ward (five counties sample)

	Dep. Var. Mean [SD]	Percent of LMCP Sites that are WB-funded	N
<i>Panel A: 2009 Census</i>			
Age 14 or Under	51.39 [3.76]	-0.01 (0.01)	170
Consumption	3063.59 [1285.98]	1.57 (3.00)	170
Primary Education	61.54 [4.54]	-0.01 (0.01)	170
Secondary Education	19.65 [6.50]	0.02 (0.02)	170
Solar Home System	1.10 [0.71]	-0.00 (0.00)	170
Electricity	6.96 [10.37]	0.04* (0.03)	170
High-Quality Wall	13.06 [9.24]	0.01 (0.03)	170
High-Quality Roof	81.52 [12.04]	-0.01 (0.03)	170
Population	22801.28 [6158.08]	5.64 (21.95)	170
Land Area (sq km)	62.70 [44.15]	0.17* (0.10)	170
<i>Panel B: Political and Ethnic Divisions</i>			
2013 Kenyatta Share (%)	33.84 [42.72]	-0.01 (0.03)	149
Voted pro-MP in 2013	0.71 [0.46]	-0.00 (0.00)	121
Luo-aligned	0.25 [0.44]	0.00 (0.00)	169
Kikuyu-aligned	0.03 [0.17]	0.00 (0.00)	169
Kalenjin-aligned	0.27 [0.45]	0.00 (0.00)	169
Joint F-test		p-value = .05	

Notes: This table tests for correlations between the share of Last Mile Connectivity Project (LMCP) sites in a ward allocated to World Bank (WB) funding and baseline characteristics, at the ward level, among wards with at least 1 study site. [Table B5](#) shows a version testing balance among all LMCP sites nationwide. Panel A tests balance on characteristics from the 2009 national census. Row 1 shows population share aged 14 years or younger. Row 2 shows monthly consumption expenditures per capita in Kenya Shillings (Ksh). Rows 3 and 4 show primary and secondary school completion rates, respectively. Rows 5–8 show percentage of households with solar, electricity, a high wall and roof quality, respectively. Rows 8–9 show the total ward population and land area, respectively. Panel B tests balance on political and ethnic data. Row 10 show the percentage of the ward that voted for Kenyatta in the 2013 presidential election. Row 11 is a dummy variable equal to one if the ward voted for the same party as their constituency MP. Rows 12–14 are dummy variables showing political alignment with major ethnic groups, equal to one if the ward voted for a member of that ethnic group in the 2013 presidential election. All socioeconomic characteristics are for the year 2009.

Table B8: Machine Learning methods to predict assignment of LMCP sites

Predictor	LPM LASSO Coefficient		Logit LASSO Coefficient		Decision Tree Importance	
	(1)	(2)	(3)	(4)	(5)	(6)
2013 Kenyatta Share	0.061	0.061	0.256	0.26	77	42
Age 14 or Under	-0.079		-0.357		63	34
Consumption	0.008		0.04		25	
Drive Distance	0.013	0	0.062		30	
Drive Time		0.001		0.003	22	
Electricity	-0.082	-0.016	-0.392	-0.082	51	
Ethnically Kalenjin-aligned	0.051	0.023	0.22	0.114	38	15
Ethnically Kikuyu-aligned	-0.002		-0.004		10	
Ethnically Luo-aligned	-0.012		-0.057		4	
High-Quality Roof	-0.034		-0.154		55	
High-Quality Wall					37	
Land Area	0.009	0.001	0.045	0.007	49	
Land Gradient	0.009	0.007	0.038	0.038	18	
Population	-0.005		-0.022		32	12
Primary Education	0.051	0.02	0.229	0.099	51	
Secondary Education	-0.008	-0.005	-0.032	-0.026	46	
Solar Home System	-0.041		-0.182	-0.004	42	10
VIIRS Radiance	0.018	0.007	0.084	0.053	33	
Voted pro-MP in 2013	-0.017	-0.011	-0.08	-0.05		
Const FEs?	No	Yes	No	Yes	No	Yes
Test RMSE	0.471	0.503	0.471	0.507	0.465	0.468
Test MAE	0.452	0.466	0.451	0.466	0.413	0.427
OoS R2	0.087	-0.043	0.088	-0.059	0.109	0.097
Class Rate	0.659	0.589	0.661	0.593	0.674	0.675

Notes: Only sites for which we have GPS coordinates and all variables (1,841 WB and 2,491 AfDB, out of a total sample of 3,308 WB and 4,184 sites). [Table B9](#) includes all sites, with missing data imputed with the mean. The classification accuracy rate of sites to funders is on average 63%: this is only slightly higher than 57%, which is what would result from simply assigning all sites to AfDB, since around 57% of sites in this sample are constructed through AfDB-funded contracts. The most predictive coefficients are baseline electricity access (in the LASSO method) and Kenyatta vote share in the 2013 presidential election (in both methods). However, the balance test discussed above detects no statistically significant differences along these dimensions ([Table B5](#)).

Table B9: Machine learning methods to predict assignment of LMCP sites

Predictor	LPM LASSO Coefficient		Logit LASSO Coefficient		Decision Tree Importance	
	(1)	(2)	(3)	(4)	(5)	(6)
2013 Kenyatta Share	0.059	0.058	0.246	0.238	107	51
Age 14 or Under	-0.051		-0.235		58	29
Consumption	0.002	-0.004	0.012	-0.016	55	
Drive Distance	0.014	0.015	0.064	0.057	29	
Drive Time		0.001		0.022	17	
Electricity	-0.068	-0.006	-0.319	-0.026	42	
Ethnically Kalenjin-aligned	0.055	0.026	0.232	0.132	73	41
Ethnically Kikuyu-aligned		0.014		0.087	14	
Ethnically Luo-aligned	-0.013		-0.06		2	
High-Quality Roof	-0.023	-0.002	-0.103	-0.011	48	17
High-Quality Wall	-0.01		-0.049		49	
Land Area			0.003		45	
Land Gradient	0.016	0.006	0.07	0.033	2	9
Population	-0.007	-0.006	-0.032	-0.032	19	
Primary Education	0.037	0.022	0.168	0.107	43	
Secondary Education		-0.005		-0.034	53	
Solar Home System	-0.033	-0.008	-0.149	-0.044	25	
VIIRS Radiance	0.018	0.009	0.086	0.072	12	
Voted pro-MP in 2013	-0.021	-0.014	-0.096	-0.067	2	
Const FEs?	No	Yes	No	Yes	No	Yes
Test RMSE	0.474	0.512	0.474	0.517	0.471	0.472
Test MAE	0.454	0.47	0.453	0.469	0.431	0.429
OoS R2	0.087	-0.066	0.088	-0.086	0.099	0.093
Class Rate	0.633	0.579	0.637	0.579	0.653	0.669

Notes: Sample: all sites for which we have GPS coordinates (2,648 WB and 3,484 AfDB). Columns 1–2 show coefficients from LASSO with a standard Linear Probability Model (LPM). Columns 3–4 show coefficients from a LASSO model with a logit link function. Columns 5–6 show variable importance from a classification tree model. Missing data imputed with the mean. [Table B8](#) only includes sites for which we have all data—results are similar.

Table B10: Heterogeneity by share of contractor's sites under audit

Outcome 4: Household installation quality index	
Treated	0.32*** (0.11)
Treated x ManyTreated	-0.27** (0.12)
Outcome 5: Household cost, experience, bribery index	
Treated	-0.04 (0.11)
Treated x ManyTreated	0.17 (0.12)
Outcome 6: Reliability and safety index	
Treated	-0.11 (0.13)
Treated x ManyTreated	0.10 (0.15)
Outcome 7: Knowledge index	
Treated	-0.04 (0.11)
Treated x ManyTreated	0.09 (0.13)
Outcome 8: Electricity Usage index	
Treated	0.31*** (0.11)
Treated x ManyTreated	-0.17 (0.12)

Test for null hypothesis of equal treatment effects for all outcomes: $p=0.252$

Notes: This table tests whether audit treatment effects differ across contractors that had a higher percentage of their sites in the treatment group. Outcome variables are indices constructed from groups of variables standardized to have mean 0 and standard deviation 1. "Treated" is a binary variable that equals 1 if the occupant is at a treated site. "ManyTreated" is a binary variable if the occupant's site is assigned to the contractor with the highest percentage of sites in the audit treatment group for its funder. All equations include constituency fixed effects, funder fixed effects, census controls, and land gradient and public facility type controls (given some baseline imbalance along these dimensions) and were jointly estimated using seemingly unrelated regression. Standard errors are clustered by site and shown in parentheses. * ≤ 0.10 , ** $\leq .05$, *** $\leq .01$.

Table B11: Impact of transformer characteristics on construction at site

	Uncompleted		
	Mean	Completed	N
World Bank (=1)	0.55 [0.50]	-0.17** (0.06)	378
Audit treatment (=1)	0.49 [0.50]	0.03 (0.06)	378
Baseline nighttime radiance	0.48 [1.03]	-0.21** (0.07)	366
Land gradient	5.55 [3.47]	-1.24*** (0.27)	347
Nearest city (KM)	70.79 [15.57]	2.13 (1.21)	366
Nearest city (minutes driving)	56.63 [28.60]	1.25 (2.02)	366
Public building...			
Health	0.08 [0.27]	-0.03 (0.02)	378
Secondary school	0.05 [0.21]	0.03 (0.03)	378
Primary school	0.16 [0.36]	0.09 (0.05)	378
Market center	0.13 [0.34]	0.01 (0.04)	378
Religious building	0.06 [0.24]	-0.03 (0.02)	378
None	0.20 [0.40]	-0.00 (0.05)	378
School	0.19 [0.39]	0.00 (0.05)	378
Other	0.14 [0.35]	-0.06 (0.04)	378
Mean	0.66	0.66	

Notes: Differences between sites that saw construction and sites that did not, among the tracked sample of 378.

Table B12: Impact of transformer characteristics on construction at site, by funder

	Bundled			Unbundled with monitoring		
	Uncompleted	Completed	N	Uncompleted	Completed	N
Baseline nighttime radiance	0.4 [0.6]	-0.1 (0.1)	187	0.5 [1.3]	-0.3* (0.1)	179
Land gradient	4.9 [2.8]	-1.5*** (0.3)	177	6.2 [3.9]	-0.8 (0.5)	170
Nearest city (KM)	73.7 [15.9]	-0.9 (1.7)	187	68.1 [14.9]	5.2** (1.9)	179
Nearest city (minutes driving)	52.1 [21.7]	-1.6 (3.0)	187	60.8 [33.3]	1.6 (3.0)	179
Public building...						
Health	0.1 [0.4]	-0.1 (0.0)	189	0.0 [0.2]	0.0 (0.0)	189
Secondary school	0.1 [0.3]	0.0 (0.1)	189	0.0 [0.1]	0.0 (0.0)	189
Primary school	0.3 [0.5]	0.1 (0.1)	189	0.0 [0.2]	-0.0 (0.0)	189
Market center	0.2 [0.4]	-0.0 (0.1)	189	0.1 [0.3]	0.1 (0.1)	189
Religious building	0.1 [0.3]	-0.1* (0.0)	189	0.0 [0.2]	-0.0 (0.0)	189
None	0.0 [0.2]	0.0 (0.0)	189	0.3 [0.5]	0.0 (0.1)	189
School	0.1 [0.3]	0.0 (0.1)	189	0.3 [0.5]	0.0 (0.1)	189
Other	0.1 [0.3]	0.0 (0.1)	189	0.2 [0.4]	-0.1 (0.1)	189
Mean	0.70	0.70		0.62	0.62	

Notes: Differences between sites that saw construction and sites that did not, among the tracked sample of 378, by whether sites saw construction under bundled contracting without additional monitoring (observed at African Development Bank sites) and under unbundled contracting with additional monitoring (observed at World Bank sites)

Table B13: Construction timing

	(1)	(2)	(3)	(4)	(5)
	AfDB Mean	β_1 : Unbundled with Additional Monitoring	β_2 : Audit Treatment (unbundled sites)	β_3 : Audit Treatment (bundled sites)	N
Outcome 3: Construction timing index	0.00 [1.00]	-0.90*** (0.17)	-0.07 (0.16)	-0.29* (0.17)	250
LMCP construction start date (months since Jan 2015)	37.22 [11.38]	10.18*** (1.90)	1.66 (1.72)	4.00** (1.93)	250
Pole erection completion date (months since Jan 2015)	45.20 [15.17]	9.90*** (2.67)	1.85 (2.49)	3.52 (2.59)	249
Stringing completion date (months since Jan 2015)	46.91 [15.48]	9.47*** (2.76)	1.33 (2.52)	2.70 (2.56)	247
Metering completion date (months since Jan 2015)	47.73 [14.56]	15.67*** (2.48)	-1.23 (2.17)	4.71* (2.65)	226
months between construction start and pole erection complete	7.83 [10.19]	-0.06 (1.81)	0.18 (1.63)	-0.32 (1.52)	249
months between pole erection complete and stringing complete	1.90 [4.41]	-0.73 (0.80)	-0.48 (0.64)	-0.53 (0.68)	246
months between stringing complete and metering complete	0.95 [8.04]	6.25*** (1.53)	-2.01* (1.20)	0.37 (1.47)	224

Notes: β_1 , β_2 , and β_3 are estimated as per Equation 1. Column (1) displays the impact of unbundled contracting and additional monitoring (observed at World Bank sites). Columns (2) and (3) display the audit treatment effect under bundled contracting without additional monitoring (observed at African Development Bank sites) and under unbundled contracting with additional monitoring (observed at World Bank sites), respectively. The construction timing index (shown here in row 1) is a standardized average of sub-components shown in the remaining rows. All outcomes are measured at the site level and collected via surveys with village representatives (described in section 4). * ≤ 0.10 , ** $\leq .05$, *** $\leq .01$.

Table B14: Impacts of contracting structure on power and voltage

	Hours of power		Average voltage	
	(1)	(2)	(3)	(4)
β_1 : Unbundled with additional monitoring	-0.19	-0.31	1.72	2.86
	(0.21)	(0.22)	(2.34)	(2.72)
β_2 : Audit treatment (unbundled sites)	-0.00	0.33*	3.45	1.39
	(0.24)	(0.17)	(2.22)	(1.77)
β_3 : Audit treatment (bundled sites)	-0.15	0.10	4.35**	4.95*
	(0.18)	(0.18)	(2.01)	(2.59)
Observations	9906	9906	654541	645648
Fixed effects	No	Yes	No	Yes
Control Mean	23.10	23.10	232.63	232.63
F-test $\beta_2 = \beta_3$ (p-val)	0.63	0.42	0.77	0.25

Notes: β_1 , β_2 , and β_3 are estimated as per [Equation 1](#) to estimate the effects of construction under bundled contracting without additional monitoring (observed at African Development Bank sites) and under unbundled contracting with additional monitoring (observed at World Bank sites). Columns (1) and (2) display daily hours of power per site. Columns (3) and (4) display hourly voltage per respondent (some observations are dropped in Column (4) in cases where there is only one observation in a fixed effects cell). Nominal voltage in Kenya is 240V. Column (2) contains week of sample by constituency fixed effects (interacted) and Column (4) contains day of sample by hour of day by constituency fixed effects (all interacted). Power quality is measured using GridWatch devices. * ≤ 0.10 , ** $\leq .05$, *** $\leq .01$.

Table B15: Primary engineering and socioeconomic outcomes with funder–audit interaction

	WB Effect Estimate	Audit Treatment Estimate	Interaction Estimate	N
Outcome 1: Construction quality index	0.64*** (0.21)	-0.03 (0.18)	0.13 (0.28)	250
Outcome 2: Network size and configuration index	-0.04 (0.16)	-0.08 (0.18)	0.27 (0.24)	244
Outcome 3: Construction timing index	-0.90*** (0.17)	-0.29* (0.17)	0.22 (0.24)	250
Outcome 4: Household installation quality index	0.05 (0.12)	0.23* (0.12)	-0.21 (0.17)	944
Outcome 5: Household cost, experience, bribery index	0.13 (0.12)	0.11 (0.10)	-0.06 (0.16)	944
Outcome 6: Reliability and safety index	-0.11 (0.13)	-0.01 (0.11)	0.04 (0.18)	944
Outcome 7: Knowledge index	0.14 (0.10)	0.07 (0.10)	-0.07 (0.14)	944
Outcome 8: Electricity Usage index	0.12 (0.13)	0.28** (0.13)	-0.17 (0.17)	944

Notes: β_1, β_2 , and β_3 are estimated as per Equation 1. Column (1) displays the impact of unbundled contracting and additional monitoring (observed at World Bank sites). Columns (2) and (3) display the audit treatment effect under bundled contracting without additional monitoring (observed at African Development Bank sites) and under unbundled contracting with additional monitoring (observed at World Bank sites), respectively. Outcome variables are indices constructed from groups of variables standardized to have mean 0 and standard deviation. In rows 1–3, observations are transformer sites; standard errors are shown in parentheses. In rows 4–8, observations are individual respondents. Table 3 presents a version with separate treatment effects. All regressions control for site land gradient and public facility type. Standard errors are clustered by transformer site and shown in parentheses. * ≤ 0.10 , ** $\leq .05$, *** $\leq .01$.

Table B16: Construction quality

	(1) AfDB Mean	(2) β_1 : Unbundled with Additional Monitoring	(3) β_2 : Audit Treatment (unbundled sites)	(4) β_3 : Audit Treatment (bundled sites)	(5) N
Outcome 1: Construction quality index	0.00 [1.00]	0.64*** (0.21)	0.10 (0.20)	-0.03 (0.18)	250
* Transformer does not have bypassed fuse	0.40 [0.49]	-0.15* (0.08)	-0.05 (0.08)	-0.08 (0.08)	250
Pole does not have a crack ≥ 1 cm	0.74 [0.44]	0.05 (0.03)	0.00 (0.03)	-0.01 (0.03)	21022
Pole leaning at ≥ 85 degrees	0.97 [0.16]	0.01* (0.00)	0.01* (0.00)	0.00 (0.00)	21229
Line has ≥ 0.5 m horiz clearance	0.93 [0.25]	-0.03*** (0.01)	0.01 (0.01)	-0.02** (0.01)	19780
Pole has cap	0.28 [0.45]	0.33*** (0.04)	0.03 (0.04)	0.06 (0.04)	17900
Stay/strut properly installed	0.92 [0.27]	0.01 (0.02)	-0.01 (0.01)	0.00 (0.02)	3193
Stay/strut installed when required	0.79 [0.41]	0.16*** (0.03)	0.02 (0.02)	0.01 (0.04)	9811
Insulator properly installed	0.99 [0.10]	-0.02* (0.01)	0.00 (0.01)	-0.00 (0.01)	3076
Insulator installed when required	0.98 [0.13]	0.01* (0.01)	-0.01* (0.01)	0.01 (0.01)	3103
Pole has grounding wire	0.34 [0.47]	0.03** (0.01)	0.01 (0.01)	-0.02* (0.01)	21229

Notes: β_1, β_2 , and β_3 are estimated as per Equation 1. Column (1) displays the impact of unbundled contracting and additional monitoring (observed at World Bank sites). Columns (2) and (3) display the audit treatment effect under bundled contracting without additional monitoring (observed at African Development Bank sites) and under unbundled contracting with additional monitoring (observed at World Bank sites), respectively. The construction quality index (shown here in row 1) is a standardized average of sub-components shown in the remaining rows. Transformer bypassed fuse is measured once at each site. All other outcomes are measured for all poles measured in the engineering assessment survey (described in Section 4). For each pole-level outcome, the sample is limited to poles for which that outcome can be assessed. Standard errors are clustered by site. An F-test of $H_0 : \beta_1 - \beta_3 = 0$ for the metering completion date has a p-val < 0.001. * ≤ 0.10 , ** $\leq .05$, *** $\leq .01$.

Table B17: Household installation quality

	(1)	(2)	(3)	(4)	(5)
	AfDB Mean	β_1 : Unbundled with Additional Monitoring	β_2 : Audit Treatment (unbundled sites)	β_3 : Audit Treatment (bundled sites)	N
Outcome 4: Household installation quality index	0.00 [1.00]	0.05 (0.12)	0.02 (0.11)	0.23* (0.12)	944
Outcome 4 (omitting readyboard question)	-0.00 [1.00]	0.15 (0.12)	-0.03 (0.12)	0.23* (0.12)	944
Electricity has flowed to this household (=1)	0.81 [0.39]	0.05 (0.06)	0.04 (0.04)	0.08 (0.05)	944
Household has ≥ 1 meter (=1)	0.86 [0.35]	0.09** (0.04)	0.01 (0.04)	0.08* (0.04)	944
Household has meter that has worked (=1)	0.77 [0.42]	0.06 (0.06)	0.07 (0.05)	0.11** (0.05)	943
Household has a readyboard (=1)	0.26 [0.44]	-0.14*** (0.04)	0.08** (0.04)	0.02 (0.05)	944
(-) Number of unrequested meters (of hhs w/ meter)	0.51 [0.50]	-0.04 (0.07)	0.10* (0.06)	0.09 (0.06)	713
(-) Weeks from paperwork to receiving meter (of hhs w/ meter)	13.64 [25.10]	4.32 (2.95)	1.58 (2.32)	-2.09 (2.47)	884
(-) Weeks from meter to receiving electricity (of hhs with elec)	2.43 [4.12]	-0.26 (0.44)	0.93* (0.54)	-0.82* (0.46)	761

Notes: β_1 , β_2 , and β_3 are estimated as per Equation 1. Column (1) displays the impact of unbundled contracting and additional monitoring (observed at World Bank sites). Columns (2) and (3) display the audit treatment effect under bundled contracting without additional monitoring (observed at African Development Bank sites) and under unbundled contracting with additional monitoring (observed at World Bank sites), respectively. The household installation quality index (shown here in rows 1 and 2) is a standardized average of sub-components shown in the remaining rows. Row 2 omits the ready board question as the absence of a ready board is not strictly an indication of poor quality. All outcomes are measured at the household level and collected in the household and firm survey data (described in Section 4). For outcomes marked with a (-), a higher value indicates a lower quality. For all other outcomes, a higher value indicates higher quality. * ≤ 0.10 , ** $\leq .05$, *** $\leq .01$.

Table B18: Electricity Usage

	(1) AfDB Mean	(2) β_1 : Unbundled with Additional Monitoring	(3) β_2 : Audit Treatment (unbundled sites)	(4) β_3 : Audit Treatment (bundled sites)	(5) N
Outcome 8: Electricity Usage index	-0.00 [1.00]	0.12 (0.12)	0.10 (0.10)	0.28** (0.12)	944
Electricity is main source of lighting (=1)	0.73 [0.44]	0.06 (0.06)	0.03 (0.05)	0.13** (0.05)	944
Electricity is main source of cooking (=1)	0.00 [0.00]	0.00 (.)	0.00 (.)	0.00 (.)	944
Household has topped up (=1) (of hh w/ prepaid meter)	0.86 [0.35]	0.02 (0.05)	0.08** (0.03)	0.11** (0.05)	836
Electricity spending past month (Ksh) (of hh w/ meter) (w)	183.13 [241.18]	-9.93 (24.36)	-0.35 (19.53)	11.54 (25.43)	893
Hours of lighting used at night in past week	2.78 [2.74]	0.10 (0.29)	0.29 (0.20)	0.40 (0.30)	848
Hours of lighting used in morning in past week	4.66 [5.69]	0.63 (0.77)	1.50** (0.74)	0.32 (0.70)	652
Number of appliances that use the grid	1.90 [1.51]	0.31* (0.17)	0.08 (0.17)	0.32** (0.16)	938
Number of households in this compound connected	1.13 [0.67]	0.01 (0.04)	0.01 (0.04)	0.03 (0.06)	944

Notes: β_1, β_2 , and β_3 are estimated as per Equation 1. Column (1) displays the impact of unbundled contracting and additional monitoring (observed at World Bank sites). Columns (2) and (3) display the audit treatment effect under bundled contracting without additional monitoring (observed at African Development Bank sites) and under unbundled contracting with additional monitoring (observed at World Bank sites), respectively. The electricity usage index (shown here in row 1) is a standardized average of sub-components shown in the remaining rows. All outcomes are measured at the household or firm level and collected in the household and firm survey data (described in Section 4). * ≤ 0.10 , ** $\leq .05$, *** $\leq .01$.

Table B19: Household cost, experience, and bribery

	(1)	(2)	(3)	(4)	(5)
	AfDB Mean	β_1 : Unbundled with Additional Monitoring	β_2 : Audit Treatment (unbundled sites)	β_3 : Audit Treatment (bundled sites)	N
Outcome 5: Household cost, experience, bribery index	-0.00 [1.00]	0.13 (0.12)	0.05 (0.11)	0.11 (0.10)	944
Days given to fulfill paperwork reqs (of LMCP hh)	42.29 [79.87]	21.09 (14.35)	0.30 (13.54)	3.16 (11.70)	828
Did not require own wiring before connection (=1)	0.77 [0.42]	-0.03 (0.05)	-0.04 (0.05)	0.01 (0.05)	855
(-) KSH spent on wiring (of hh that did wiring) (w)	7774.45 [6779.96]	-925.05 (718.32)	645.25 (666.29)	-741.25 (739.09)	708
(-) Up-front connection payment (Ksh) (w)	6684.48 [9104.41]	-694.60 (844.78)	588.85 (776.80)	-685.49 (923.51)	925
Connected by KPLC/REA (=1)	0.98 [0.13]	0.01 (0.02)	0.01 (0.01)	0.01 (0.01)	837
Was not asked for bribe (=1)	0.91 [0.29]	0.02 (0.03)	-0.02 (0.03)	-0.01 (0.03)	944
Didn't do unpaid manual labor for connection (=1)	0.96 [0.19]	-0.02 (0.02)	0.04** (0.02)	0.00 (0.02)	929
(-) Amount paid so far in installments (Ksh) (w)	2698.65 [4531.45]	-24.92 (521.88)	-454.06 (467.42)	-48.46 (504.09)	878
Satisfaction with electricity installation (1-5 scale)	4.21 [1.07]	-0.02 (0.13)	0.04 (0.12)	0.08 (0.13)	944
(-) Hours in past month with very low voltage	1.57 [6.61]	2.85 (1.86)	1.07 (1.73)	-1.80 (1.67)	602
(-) Repair costs for devices damaged b/c electricity (Ksh)	31.19 [206.11]	-9.37 (32.01)	-44.27** (22.40)	-67.32** (33.07)	604

Notes: β_1, β_2 , and β_3 are estimated as per [Equation 1](#). Column (1) displays the impact of unbundled contracting and additional monitoring (observed at World Bank sites). Columns (2) and (3) display the audit treatment effect under bundled contracting without additional monitoring (observed at African Development Bank sites) and under unbundled contracting with additional monitoring (observed at World Bank sites), respectively. The household cost, experience, and bribery index (shown here in row 1) is a standardized average of sub-components shown in the remaining rows. All outcomes are measured at the household or firm level and collected in the household and firm survey data (described in [Section 4](#)). For outcomes marked with a (-), a higher value indicates a lower quality. For all other outcomes, a higher value indicates higher quality.

* ≤ 0.10 , ** ≤ 0.05 , *** ≤ 0.01 .

Table B20: Household and firm reliability and safety

	(1)	(2)	(3)	(4)	(5)
	AfDB Mean	β_1 : Unbundled with Additional Monitoring	β_2 : Audit Treatment (unbundled sites)	β_3 : Audit Treatment (bundled sites)	N
Outcome 6: Reliability and safety index	-0.00 [1.00]	-0.11 (0.13)	0.02 (0.14)	-0.02 (0.11)	944
Had power in past 7 days (=1) (of electrified hh)	0.88 [0.32]	0.06 (0.04)	-0.02 (0.04)	0.11*** (0.03)	787
No regular blackouts (=1) (of electrified hh)	0.58 [0.49]	-0.11** (0.06)	0.03 (0.06)	-0.05 (0.06)	787
No blackout in past 7 days (=1) (of hh w/ power last 7 days)	0.40 [0.49]	0.01 (0.07)	0.01 (0.07)	0.07 (0.07)	703
(-) Hours power not working in past 7 days (of hh w/ power last 7 days)	7.12 [15.04]	1.74 (1.91)	-2.86* (1.66)	0.56 (1.86)	700
No blackouts ≥ 30 days in past year (=1) (of electrified hh)	0.95 [0.23]	-0.06 (0.04)	0.01 (0.04)	-0.02 (0.03)	787
No injury fr/ electricity in past year (=1) (of electrified hh)	0.99 [0.10]	0.00 (0.01)	-0.02 (0.01)	-0.01 (0.01)	787
No damage fr/ electricity in past year (=1) (of electrified hh)	0.99 [0.09]	-0.01 (0.01)	0.00 (0.01)	-0.02** (0.01)	787

Notes: β_1 , β_2 , and β_3 are estimated as per Equation 1. Column (1) displays the impact of unbundled contracting and additional monitoring (observed at World Bank sites). Columns (2) and (3) display the audit treatment effect under bundled contracting without additional monitoring (observed at African Development Bank sites) and under unbundled contracting with additional monitoring (observed at World Bank sites), respectively. The household reliability and safety index (shown here in row 1) is a standardized average of sub-components shown in the remaining rows. All outcomes are measured at the household or firm level and collected in the household and firm survey data (described in Section 4). For outcomes marked with a (-), a higher value indicates a lower quality. For all other outcomes, a higher value indicates higher quality. * ≤ 0.10 , ** $\leq .05$, *** $\leq .01$.

Table B21: Knowledge

	(1)	(2)	(3)	(4)	(5)
	AfDB Mean	β_1 : Unbundled with Additional Monitoring	β_2 : Audit Treatment (unbundled sites)	β_3 : Audit Treatment (bundled sites)	N
Outcome 7: Knowledge index	0.00 [1.00]	0.13 (0.09)	-0.02 (0.09)	0.06 (0.10)	944
Told correct total cost of connection (=1) (of hh w/ drop cable)	0.29 [0.46]	0.05 (0.06)	0.02 (0.07)	0.02 (0.06)	930
Correctly told to pay monthly (=1) (of hh told of connxn cost)	0.05 [0.22]	-0.05*** (0.02)	0.02 (0.01)	0.00 (0.02)	930
Knows how much still owed for connection (=1)	0.43 [0.50]	0.16*** (0.06)	-0.07 (0.06)	0.02 (0.06)	944
Knows 20th token costs same as 1st (=1) (of hh who have topped up)	0.76 [0.43]	0.02 (0.06)	-0.02 (0.07)	-0.01 (0.06)	707
Knows value of 1st token	0.94 [0.23]	0.01 (0.03)	-0.00 (0.02)	0.02 (0.03)	707

Notes: β_1, β_2 , and β_3 are estimated as per Equation 1. Column (1) displays the impact of unbundled contracting and additional monitoring (observed at World Bank sites). Columns (2) and (3) display the audit treatment effect under bundled contracting without additional monitoring (observed at African Development Bank sites) and under unbundled contracting with additional monitoring (observed at World Bank sites), respectively. The knowledge index (shown here in row 1) is a standardized average of sub-components shown in the remaining rows. All outcomes are measured at the household or firm level and collected in the household and firm survey data (described in Section 4). * ≤ 0.10 , ** $\leq .05$, *** $\leq .01$.

Table B22: Network configuration

	(1)	(2)	(3)	(4)	(5)
	AfDB Mean	β_1 : Unbundled with Additional Monitoring	β_2 : Audit Treatment (unbundled sites)	β_3 : Audit Treatment (bundled sites)	N
Outcome 2: Network size and configuration index	0.00 [1.00]	-0.04 (0.16)	0.19 (0.16)	-0.08 (0.18)	244
Absolute Deviation in Pole Count (relative to design)	65.66 [55.90]	-3.87 (11.44)	6.44 (11.26)	2.46 (12.42)	197
Absolute Deviation in Drop Cables (relative to design)	62.12 [44.79]	15.69 (9.57)	-0.07 (9.49)	11.78 (10.49)	178
Fraction of compounds at site, within 100m of LV line, electrified	0.89 [0.13]	-0.02 (0.02)	0.04 (0.03)	-0.01 (0.02)	244
Fraction of poles ≤ 600 m from transformer	0.95 [0.08]	0.02 (0.01)	0.01 (0.01)	0.00 (0.01)	244
Number of poles in design	134.51 [87.55]	-1.10 (15.32)	-8.89 (16.60)	12.52 (15.92)	197

Notes: β_1, β_2 , and β_3 are estimated as per Equation 1. Column (1) displays the impact of unbundled contracting and additional monitoring (observed at World Bank sites). Columns (2) and (3) display the audit treatment effect under bundled contracting without additional monitoring (observed at African Development Bank sites) and under unbundled contracting with additional monitoring (observed at World Bank sites), respectively. The network size and configuration index (shown here in row 1) is a standardized average of sub-components shown in the remaining rows. All outcomes are measured at the household or firm level and collected in the household and firm survey data (described in Section 4). * ≤ 0.10 , ** $\leq .05$, *** $\leq .01$.

Table B23: Designs

	(1)	(2)	(3)	(4)	(5)
	AfDB Mean	World Bank Effect Estimate (β_1)	Audit Treatment Effect, WB Sites (β_2)	Audit Treatment Effect, AfDB Sites (β_3)	N
Number of poles in site	111.66 [63.72]	4.44 (12.73)	-16.48 (13.42)	9.18 (14.01)	179
Distance pole to nearest pole (m)	37.98 [6.21]	-1.42 (1.63)	0.58 (1.91)	0.43 (1.66)	176
Fraction of poles outside 600m boundary	0.14 [0.09]	0.00 (0.04)	-0.03 (0.03)	-0.03 (0.03)	179
Pole height (m)	9.83 [0.22]	-0.03 (0.05)	0.05 (0.05)	-0.04 (0.05)	179
Number of meters in site	21.52 [37.11]	-3.46 (8.42)	-7.58 (8.54)	-6.80 (8.24)	179
Distance meter to nearest pole (m)	16.87 [7.17]	0.53 (2.06)	1.03 (1.49)	3.37 (2.05)	156
Fraction of meters outside 600m boundary	0.13 [0.21]	-0.05 (0.05)	-0.07* (0.04)	0.01 (0.05)	156

Notes: β_1 , β_2 , and β_3 are estimated as per [Equation 1](#). Column (1) displays the impact of unbundled contracting and additional monitoring (observed at World Bank sites). Columns (2) and (3) display the audit treatment effect under bundled contracting without additional monitoring (observed at African Development Bank sites) and under unbundled contracting with additional monitoring (observed at World Bank sites), respectively. Impacts on low-voltage network design features taken from the designs produced by contractors. * ≤ 0.10 , ** $\leq .05$, *** $\leq .01$.

Table B24: Household Socioeconomic Outcomes

	(1)	(2)	(3)	(4)	(5)
	AfDB Mean	β_1 : Unbundled with Additional Monitoring	β_2 : Audit Treatment (unbundled sites)	β_3 : Audit Treatment (bundled sites)	N
Outcome 9: Household socioeconomic outcomes index	-0.00 [1.00]	0.24* (0.12)	-0.01 (0.13)	0.20 (0.13)	944
Connection allowed pursuing employment, business (1-5) (of connected hh)	2.54 [1.19]	0.27* (0.15)	0.33** (0.14)	0.16 (0.15)	787
Connection affected earnings (1-5) (of connected hh)	3.25 [0.78]	0.15* (0.09)	0.09 (0.09)	0.01 (0.09)	787
Connection permitted changing hours worked (1-5) (of connected hh)	3.65 [0.86]	0.05 (0.12)	0.05 (0.12)	0.04 (0.11)	787
Connection affected amount of food consumed (1-5) (of connected hh)	3.10 [0.45]	0.14** (0.05)	0.03 (0.05)	0.08 (0.06)	787
Connection affected health (1-5) (of connected hh)	3.59 [0.86]	-0.08 (0.11)	0.08 (0.10)	-0.05 (0.11)	787
Connection affected children's education (1-5) (of connected hh w/ children)	4.32 [0.85]	0.33*** (0.09)	-0.04 (0.08)	0.19* (0.10)	691
Connection affected knowledge about news (1-5) (of connected hh)	4.15 [0.97]	0.14 (0.10)	0.01 (0.09)	0.10 (0.10)	787
Connection permitted changing kerosene spending (1-5) (of connected hh)	1.51 [0.99]	-0.03 (0.10)	0.06 (0.10)	0.07 (0.10)	787
Connection changed phone charging freq. (1-5) (of connected hh)	3.11 [1.49]	0.57*** (0.18)	-0.13 (0.19)	0.36** (0.17)	787
(-) Kerosene spending, last week (Ksh) (w)	30.02 [62.30]	-15.21** (6.04)	15.52** (6.32)	-8.91 (5.80)	940
Owns home (=1)	0.99 [0.10]	0.00 (0.01)	-0.01 (0.01)	0.00 (0.01)	944
Number of rooms in primary residence	3.54 [1.66]	-0.19 (0.15)	-0.05 (0.13)	0.08 (0.14)	944
High-quality floors (=1)	0.38 [0.48]	0.04 (0.05)	-0.12*** (0.05)	-0.02 (0.05)	944
High-quality roof (=1)	1.00 [0.06]	-0.01* (0.01)	0.00 (0.01)	0.01 (0.01)	944
High-quality walls (=1)	0.21 [0.41]	0.01 (0.04)	-0.00 (0.04)	0.06 (0.04)	944
Buildings in compound (of compounds with hh)	2.94 [1.56]	-0.15 (0.15)	-0.01 (0.13)	-0.18 (0.20)	747
Electrified buildings in compound (of compounds with hh)	1.64 [1.31]	-0.04 (0.10)	0.01 (0.08)	0.14 (0.16)	747

Notes: β_1 , β_2 , and β_3 are estimated as per Equation 1. Column (1) displays the impact of unbundled contracting and additional monitoring (observed at World Bank sites). Columns (2) and (3) display the audit treatment effect under bundled contracting without additional monitoring (observed at African Development Bank sites) and under unbundled contracting with additional monitoring (observed at World Bank sites), respectively. The household socioeconomic outcomes index (shown here in row 1) is a standardized average of sub-components shown in the remaining rows. All outcomes are measured at the household or firm level and collected in the household and firm survey data (described in Section 4). For outcomes marked with a (-), a higher value indicates a lower quality. For all other outcomes, a higher value indicates higher quality. Due to ambiguity in the wording for one of the survey questions, a pre-specified outcome ("connection affected security")

Table B25: Firm Performance

	(1)	(2)	(3)	(4)	(5)
	β_1 : Unbundled AfDB with Mean	Additional Monitoring	β_2 : Audit Treatment (unbundled sites)	β_3 : Audit Treatment (bundled sites)	N
Outcome 10: Firm Performance Index	-0.00 [1.00]	0.29 (0.19)	-0.11 (0.21)	0.12 (0.17)	373
Firm uses electricity (=1)	0.64 [0.48]	0.20** (0.08)	0.02 (0.08)	0.11 (0.08)	339
Firm planning to buy electrical equipment in next year (=1)	0.42 [0.49]	0.13 (0.10)	-0.11 (0.09)	0.06 (0.08)	339
Firm uses elec beyond lighting and cell charge (=1) (of those that use elec)	0.36 [0.48]	-0.08 (0.09)	0.00 (0.08)	-0.19** (0.07)	344
Number of appliances owned by Firm	1.23 [1.13]	0.24 (0.24)	-0.13 (0.24)	0.03 (0.20)	344
Firm household has high quality roof (=1)	0.89 [0.31]	0.07 (0.06)	-0.08 (0.06)	0.03 (0.06)	306
Firm household has high quality walls (=1)	0.49 [0.50]	-0.04 (0.12)	0.04 (0.10)	0.11 (0.10)	306

Notes: β_1 , β_2 , and β_3 are estimated as per [Equation 1](#). Column (1) displays the impact of unbundled contracting and additional monitoring (observed at World Bank sites). Columns (2) and (3) display the audit treatment effect under bundled contracting without additional monitoring (observed at African Development Bank sites) and under unbundled contracting with additional monitoring (observed at World Bank sites), respectively. The firm performance index (shown here in row 1) is a standardized average of sub-components shown in the remaining rows. All outcomes are measured at the firm level and collected in the household and firm survey data (described in [Section 4](#)). * ≤ 0.10 , ** $\leq .05$, *** $\leq .01$.

Table B26: Household Political and Social Beliefs

	(1)	(2)	(3)	(4)	(5)
	AfDB Mean	β_1 : Unbundled with Additional Monitoring	β_2 : Audit Treatment (unbundled sites)	β_3 : Audit Treatment (bundled sites)	N
Outcome 11: Political and Social Beliefs index	-0.00 [1.00]	0.03 (0.08)	0.01 (0.08)	0.03 (0.09)	944
HH electrification in top 2 most-important govt policies (=1)	0.21 [0.41]	0.00 (0.04)	-0.01 (0.04)	-0.01 (0.04)	944
Thinks govt doing good job providing electricity (=1)	0.98 [0.14]	0.00 (0.01)	-0.00 (0.01)	-0.01 (0.01)	944
Voted in August 2017 election (=1)	1.15 [4.42]	0.07 (0.20)	0.35 (0.33)	0.48 (0.35)	944

Notes: β_1 , β_2 , and β_3 are estimated as per Equation 1. Column (1) displays the impact of unbundled contracting and additional monitoring (observed at World Bank sites). Columns (2) and (3) display the audit treatment effect under bundled contracting without additional monitoring (observed at African Development Bank sites) and under unbundled contracting with additional monitoring (observed at World Bank sites), respectively. The household political and social beliefs index (shown here in row 1) is a standardized average of sub-components shown in the remaining rows. All outcomes are measured at the household or firm level and collected in the household and firm survey data (described in Section 4). * ≤ 0.10 , ** $\leq .05$, *** $\leq .01$.

Table B27: Audit treatment effect heterogeneity

	(1)	(2)	(3)	(4)	(5)	(6)
			Government	Gradient	Drive time	Radiance
Audit treatment (=1)	0.13 (0.08)					
Treatment X WB		0.01 (0.11)				
Treatment X AfDB		0.25* (0.13)				
Treatment X Below median			0.07 (0.09)	0.10 (0.10)	0.11 (0.10)	0.14 (0.14)
Treatment X Above median			0.19 (0.15)	0.16 (0.14)	0.15 (0.14)	0.12 (0.10)
Observations	944	944	944	944	944	944

Table B28: Impact of gradient and facility type on construction delays

Panel A) Months to stringing completion

	(1)	(2)	(3)	(4)	(5)
World Bank (=1)	6.8*** (2.1)	9.9*** (2.2)	9.5*** (2.3)	9.5*** (2.3)	8.7*** (2.5)
Land gradient			0.6 (0.6)		0.4 (0.7)
Health center				-0.3 (5.4)	1.1 (5.7)
Secondary school				-0.4 (3.3)	-1.3 (3.4)
Primary school				1.8 (2.4)	2.6 (2.6)
Market center				1.1 (2.7)	1.9 (2.9)
Religious building				-3.9 (2.9)	-4.0 (3.0)
Other				2.7 (5.8)	4.9 (6.3)
Observations	246	246	229	226	211
Constituency FE	No	Yes	Yes	Yes	Yes

Panel B) Months to metering completion

	(1)	(2)	(3)	(4)	(5)
World Bank (=1)	9.6*** (1.8)	12.4*** (1.8)	11.7*** (1.9)	13.2*** (1.9)	12.2*** (2.0)
Land gradient			1.0* (0.5)		0.8 (0.6)
Health center				3.7 (4.5)	5.3 (4.6)
Secondary school				0.9 (2.7)	0.4 (2.7)
Primary school				1.3 (2.0)	1.6 (2.1)
Market center				-2.0 (2.2)	-1.1 (2.3)
Religious building				1.4 (2.4)	1.3 (2.5)
Other				3.8 (4.7)	6.5 (5.1)
Observations	248	248	231	227	212
Constituency FE	No	Yes	Yes	Yes	Yes

Notes: Stringing (metering) was completed at WB sites on average 6.8 (9.6) months later than at AfDB sites when pooling audit control and treatment sites. Controlling for land gradient and facility type does not affect these estimates meaningfully, and land gradient and facility type appear largely uncorrelated with time to stringing and metering completion.

Table B29: Heterogeneity in WB delay by facility type

	Time to stringing completion (months)				
	(1)	(2)	(3)	(4)	(5)
World Bank (=1)	18.5 (13.5)	5.2 (6.9)	8.8* (5.0)	-6.0 (5.3)	-3.0 (14.6)
Observations	9	64	53	17	21
Control Mean	41.5	53.16	50.52	43.1	54.36
Sample	Health centers	Schools	Market centers	Religious buildings	Others

Notes: While there are small differences between funder type in the facility type associated with each transformer ([Table B4](#)) this does not drive heterogeneity in the impact of WB conditionality on construction delays when compared with AfDB sites.

* ≤ 0.10 , ** $\leq .05$, *** $\leq .01$.

Table B30: Primary engineering and socioeconomic outcomes excluding Lots 3 and 5

	(1) β_1 : Unbundled with Additional Monitoring	(2) β_2 : Audit Treatment (unbundled sites)	(3) β_3 : Audit Treatment (bundled sites)	(4) N
Outcome 1: Construction quality index	0.57 (0.50)	0.13 (0.49)	0.05 (0.19)	161
Outcome 2: Network size and configuration index	0.00 [1.00]	-0.16 (0.37)	-0.07 (0.20)	-0.07 (0.20)
Outcome 3: Construction timing index	0.00 [1.00]	-0.15 (0.39)	-0.36** (0.17)	-0.36** (0.17)
Outcome 4: Household installation quality index	0.00 [1.00]	0.79*** (0.23)	0.23* (0.13)	0.23* (0.13)
Outcome 5: Household cost, experience, bribery index	-0.00 [1.00]	0.10 (0.18)	0.11 (0.10)	0.11 (0.10)
Outcome 6: Reliability and safety index	-0.00 [1.00]	-0.18 (0.17)	-0.09 (0.10)	-0.09 (0.10)
Outcome 7: Knowledge index	0.00 [1.00]	-0.09 (0.20)	0.10 (0.10)	0.10 (0.10)
Outcome 8: Electricity Usage index	-0.00 [1.00]	0.53*** (0.19)	0.25* (0.14)	0.25* (0.14)
Outcome 10: Firm Performance Index	-0.00 [1.00]	0.08 (0.43)	0.02 (0.17)	0.02 (0.17)
Outcome 11: Political and Social Beliefs index	-0.00 [1.00]	-0.02 (0.13)	0.02 (0.09)	0.02 (0.09)

Notes: This table replicates [Table 3](#) but excludes Lots 3 and 5 to account for a legal case (see [AEE Power SA v Kenya Power & Lighting Company Ltd \(2020\)](#)) and then retains only a balanced panel of constituencies. β_1, β_2 , and β_3 are estimated as per [Equation 1](#). Column (1) displays the impact of unbundled contracting and additional monitoring (observed at World Bank sites). Columns (2) and (3) display the audit treatment effect under bundled contracting without additional monitoring (observed at African Development Bank sites) and under unbundled contracting with additional monitoring (observed at World Bank sites), respectively. [Subsection 6.5](#) provides more detail. * ≤ 0.10 , ** $\leq .05$, *** $\leq .01$.

Table B31: Connections and poles installed per site excluding nearby sites

	Entire site				Outside 600m boundary			
	Poles		Meters		Poles		Meters	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
β_1 : Unbundled with additional monitoring	-12.8**	-5.5	-13.4**	-22.2*	-2.4***	-1.7	-1.6***	-1.4
	(6.2)	(10.7)	(6.7)	(11.5)	(0.7)	(1.2)	(0.6)	(1.0)
Audit treatment	9.0		6.7		0.2		0.1	
	(6.2)		(6.6)		(0.7)		(0.6)	
β_2 : Audit treatment (unbundled sites)		-2.4		8.6		-0.6		-0.4
		(9.2)		(9.9)		(1.0)		(0.8)
β_3 : Audit treatment (bundled sites)		18.7**		4.3		0.9		0.7
		(8.8)		(9.5)		(1.0)		(0.8)
Observations	224	224	224	224	218	218	218	218
Control Mean	93.33	93.33	73.30	73.30	3.77	3.77	2.98	2.98

Notes: This table replicates [Table 2](#) but excluding sites that are less than 1,200 of another site, as the areas within 600 meters of such sites would overlap (see [Section 4](#) for a discussion of this problem). If anything, this version more strongly supports our results. β_1, β_2 , and β_3 are estimated as per [Equation 1](#). β_1 captures the impact of unbundled contracting and additional monitoring (observed at World Bank sites). β_2 and β_3 capture the impact of the audit treatment effect under bundled contracting without additional monitoring (observed at African Development Bank sites) and under unbundled contracting with additional monitoring (observed at World Bank sites), respectively. All regressions include constituency fixed effects. Standard errors shown in parentheses. * ≤ 0.10 , ** $\leq .05$, *** $\leq .01$.

Table B32: Site, connection, and materials costs by contracting approach

		Bundled contracts, no monitoring	Unbundling, additional monitoring	Percent Difference
(1)	Total contracted value (USD millions)	154	127	−18%
(2)	Sites planned	5,213	4,205	−19%
(3)	Contract amount per planned site (USD)	29,532	30,291	+3%
(4)	Observed new household connections per site	80	80	+0%
(5)	Observed new household connections per site	72	58	−19%
(6)	Sites with any construction	4,713	3,250	−31%
(7)	Contract amount per site with construction (USD)	32,668	39,187	+20%
(8)	Contract amount per household connection (USD)	454	676	+49%
(9)	Sites with complete construction	3,550	2,813	−21%
(10)	Contract amount completed site (USD)	43,366	45,278	+4%
(11)	Contract amount per household connection (USD)	602	781	+30%
(12)	Contract amount per wooden pole (USD)	159	99	−38%
(13)	Contract amount per concrete pole (USD)	240	199	−17%

Notes: Aggregate connection and pole procurement quantities and costs, per the contracts signed between Kenya Power and contractors under bundled contracting without additional monitoring (observed at African Development Bank sites) and under unbundled contracting with additional monitoring (observed at World Bank sites). Rows 1, 12, and 13 use contract data. Row 2 uses public data. Rows 3 through 11 are calculated using observed construction rates.

C.1 Qualitative interviews

From June 2016 through July 2022, members of the research team met extensively with dozens of key Kenya Power personnel. This included the General Manager for Connectivity, who was responsible for all of Kenya Power’s new electricity connections, his key managing staff; two Project Managers who oversaw the nationwide LMCP; and senior managers responsible for the WB and AfDB contracting ([Subsection C.1](#) describes a full list of qualitative interviews conducted). For both the WB and AfDB components, Kenya Power employed one full-time staff member to manage the contracting components and a handful of employees to oversee implementation. The staff who held these positions were all certified electrical engineers with similar skill and education levels, all holding at least a bachelor’s degree in electrical engineering.

Qualitative research included detailed in-person (or on Zoom, where required due to Covid-19) conversations with key leadership personnel at Kenya Power, World Bank, African Development Bank, and the Consultant charged with supervising construction. An asterisk (*) indicates that a single position was held by different individuals at different points in time.

- World Bank employees:
 - Practice manager, Global energy and extractives practice, Africa region
 - Senior energy specialist, Kenya country team
 - Energy finance specialist, Kenya country team
- African Development Bank employees:
 - Principal power engineer*
 - Principal power engineer*
- Kenya Power employees:
 - General manager of connectivity
 - General manager of infrastructure development
 - LMCP Contract Project Manager (AfDB Phase I)
 - LMCP Project Leader (AfDB Phase I)
 - LMCP Contract Project Manager (WB)
 - LMCP Project Leader (WB)
 - LMCP Project Leader for (AfDB Phase II)
- Project Management Consultant employees:
 - Senior Manager

C.2 Rural electrification in Kenya

In 2014, Kenya’s Ministry of Energy and Petroleum (MoE) published the Draft National Energy Policy, establishing a list of policies and strategies to “*increase rural electrification connectivity to at least 40% by 2016 and 100% by 2020*” and to “*seek funding from development partners for specific programmes especially...in rural electrification projects.*” ([MoE 2014](#)). In Kenya Power’s 2014-2015 annual report, they note that “*The KShs 4 Billion receivable from the GoK is part of a larger commitment by the GoK, to be financed partly*

C Additional background information

Table C1: Representative examples of contract language

World Bank	African Development Bank
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<i>Additional evaluation criteria</i>	
[2.17 ^a] The bidding documents shall specify any factors, in addition to price, which will be taken into account in evaluating bids, and how such factors will be quantified or otherwise evaluated. If bids based on alternative designs, materials, completion schedules, payment terms, etc. are permitted, conditions for their acceptability and the method of their evaluation shall be expressly stated.	[2.17 ^b] The bidding documents shall specify any factors, in addition to price, which will be taken into account in evaluating bids, and how such factors will be quantified or otherwise evaluated. If bids based on alternative designs, materials, completion schedules, payment terms, etc., are permitted, conditions for their acceptability and the method of their evaluation shall be expressly stated.
<i>Bidder cost calculations</i>	
[2.21 ^a] Bids for goods shall be invited on the basis of CIP (place of destination) for all goods manufactured abroad and to be imported. Bids for goods that were previously imported shall be invited on the basis of CIP (place of destination) separately indicating the actual amount of customs duties and import taxes already paid. Bids for goods manufactured in the Borrower's country shall be invited on the basis of EXW (ex works, ex factory, or off-the-shelf) plus cost of inland transportation and insurance to the place of destination. Bidders shall be allowed to arrange for ocean and other transportation and related insurance from any eligible source. Where installation, commissioning, or other similar services are required to be performed by the bidder, as in the case of supply and installation contracts, the bidder shall be required to quote for these services.	[2.21 ^b] Bids for goods shall be invited on the basis of CIP (place of destination) for all goods manufactured abroad, including those previously imported, and EXW (Ex works, Ex factory, or Off-the-shelf) plus cost of inland transportation and insurance to the place of destination for goods manufactured or assembled in the country of the Borrower. Bidders shall be allowed to arrange for ocean and other transportation and related insurance from any eligible source. Where installation, commissioning, or other similar services are required to be performed by the bidder, as in the case of "supply and installation" contracts, the bidder shall be required to quote for these services, in addition.
<i>Schedule of payment</i>	
[2.34 ^a] [...] (a) Contracts for supply of goods shall provide for full payment on the delivery and inspection, if so required, of the contracted goods except for contracts involving installation and commissioning, in which case a portion of the payment may be made after the Supplier has complied with all its obligations under the contract. The Bank normally requires the use of letters of credit so as to assure prompt payment to the Supplier. In major contracts for equipment and plant, provisions shall be made for suitable advances and, in contracts of long duration, for progress payments during the period of manufacture or assembly. (b) Contracts for works shall provide, in appropriate cases for mobilization advances, advances on Contractor's equipment and materials, regular progress payments, and reasonable retention amounts to be released upon compliance with the Contractor's obligations under contract.	[2.34 ^b] [...] (a) Contracts for supply of goods shall provide for full payment on the delivery and inspection, if so required, of the contracted goods except for contracts involving installation and commissioning, in which case a portion of the payment may be made after the Supplier has complied with all its obligations under the contract. The use of letters of credit is encouraged so as to assure prompt payment to the supplier. In major contracts for equipment and plant, provision shall be made for suitable advances and, in contracts of long duration, for progress payments during the period of manufacture or assembly. (b) Contracts for works shall provide in appropriate cases for mobilization advances, advances on contractor's equipment and materials, regular progress payments, and reasonable retention amounts to be released upon compliance with the Contractor's obligations under contract.
<i>Bid securities</i>	
[2.35 ^a] Contracts for supply of goods shall provide for full payment on the delivery and inspection, if so required, of the contracted goods except for contracts involving installation and commissioning, in which case a portion of the payment may be made after the Supplier has complied with all its obligations under the contract. The Bank normally requires the use of letters of credit so as to assure prompt payment to the Supplier. In major contracts for equipment and plant, provisions shall be made for suitable advances and, in contracts of long duration, for progress payments during the period of manufacture or assembly. (b) Contracts for works shall provide, in appropriate cases for mobilization advances, advances on Contractor's equipment and materials, regular progress payments, and reasonable retention amounts to be released upon compliance with the Contractor's obligations under contract.	[2.35 ^b] Bidding documents for supply of goods shall provide for full payment on the delivery and inspection, if so required, of the contracted goods except for contracts involving installation and commissioning, in which case a portion of the payment may be made after the Supplier has complied with all its obligations under the contract. The use of letters of credit is encouraged so as to assure prompt payment to the supplier. In major contracts for equipment and plant, provision shall be made for suitable advances and, in contracts of long duration, for progress payments during the period of manufacture or assembly. (b) Contracts for works shall provide in appropriate cases for mobilization advances, advances on contractor's equipment and materials, regular progress payments, and reasonable retention amounts to be released upon compliance with the Contractor's obligations under contract.

through support from the World Bank and the African Development Bank to enhance universal access to electricity.” In May 2015, Kenya’s President Uhuru Kenyatta announced the launch of the LMCP, with a goal of connecting “one million new customers to electricity each year” (Kenya Presidency, 2015). In a press conference two weeks after President Kenyatta’s announcement, Kenya Power’s then- Managing Director Ben Chumo added that the program was designed to facilitate “the government’s objective of providing 70% households with electricity by 2017 and universal access by 2020” (Kenya Power, 2015b).⁴ While not quite reaching these ambitious targets, the program has been effective: nationwide household electricity access was reported to have increased from 25% in 2009 to 70% in 2019 (KNBS 2009, 2019). Many of the rural transformers selected for the LMCP had been constructed between 2005 and 2013 as part of a nationwide push by Kenya’s then- Rural Electrification Authority (REA)⁵ to connect all public facilities—such as markets, schools, health centers, and water points—to electricity (REA 2008, Berkouwer, Lee, and Walker, 2018).

In November 2017 the AfDB signed 15 additional turn-key contracts to begin maximization of an additional 5,200 sites as part of its *Phase II* (which we do not examine in this study).

C.3 Upfront connection costs

Beneficiaries under the LMCP are connected via ‘pre-paid’ meters, meaning they must buy electricity credits in advance of using electricity. Once they consume all of their prepaid electricity, they lose access to electricity, and only regain access only after they buy more credits. Households usually prevent this by purchasing additional credits before their credits run out.

To recover the \$150 connection fee, Kenya Power initially enrolled households into a payment plan consisting of 36 monthly installments of around \$4 per month. The charge was automatically added to households’ accounts on a monthly basis, and any electricity payments the household made were directed towards paying off this debt prior to being directed towards electricity credits. However, this generated a significant barrier for households: as an example, if a household runs out of electricity credit in January, and then does not consume any electricity in February or March, they would have to pay at least \$16.01—4 months worth of connection fees—to be able to consume any electricity in April. The contribution was thus later capped at 50% of any topup amount (Kassem, Zane, and Uzor, 2026).

This barrier was not only a significant financial hurdle, but one that was unanticipated and poorly understood. According to Kenya Power, households should have been informed of the payment structure as part of the consent process, which was the very first step in the construction process, but it is unclear whether this consent process was regularly implemented in practice. To verify whether this process was correctly implemented, and to test whether donor conditionality and monitoring can improve adherence to these guidelines, the household survey (described in Section 4) measures respondent understanding of the aggregate costs of an electricity connection under the LMCP. 58% of households do not recall ever having been told that they would have to pay Kenya Power for the connection.

An additional financial hurdle was the upfront cost of wiring, which the LMCP later

⁴ This target date was later extended to 2022, which was also not met.

⁵ Since renamed Rural Electrification and Renewable Energy Corporation (REREC).

tried to address by providing ready boards. In a May 2015 address, President Kenyatta described this policy as follows: *“The Ministry of Energy has also come up with designs that will enable households that do not have internal wiring in their houses to use electricity by providing a ‘ready board’... [it] has switches, sockets and bulb holders and those who do not have wiring in their houses will be able to use electricity as soon as they are connected”* (Kenya Presidency, [2015](#)).

C.4 Informal and illegal connections

Illegal connections are much more common in urban areas than they are in rural areas like the villages where the LMCP was implemented. Many households in urban contexts, especially those living in informal settlement areas, are sufficiently close to the existing grid that they can be connected via a simple drop cable, which can usually be done by a local handyman at relatively low cost. Given the low population density in rural areas, connection of an additional household usually requires constructing at least one additional electricity pole, which requires more sophisticated engineering techniques. In our survey, only 2.7% of households with a working electricity connection did not have a meter. Of these, 93% said they had not been metered yet but would be metered soon, and 20% said they had not yet done the internal wiring that was required prior to connection. Nobody stated the reason they did not have a meter was because theirs was an illegal connection. Of course, these survey responses come with the usual caveats about survey questions relating to illegal behavior

C.5 Unconnected households

The LMCP’s objective was to connect all unconnected households to electricity, however, in practice connectivity was not universal. At the average site at least 7% of compounds were not connected to the grid, and at the 90th percentile site at least 25% of households were not connected.⁶ The most common reason (given by 31% of unconnected respondents) is that they were not present or available during the days on which construction or sign-up were administered. Second, even though the LMCP program specifications indicate there were to be no upfront connection fees, 23% of respondents still report having been unable to pay, often because they were not able to afford the internal wiring required by Kenya Power to be connected: 16% of unconnected households report this to be the reason. This suggests that despite efforts to provide free ready boards to low-income households, the cost of household wiring remained a barrier that prevented some households from getting connected.

Households could choose not to get connected, but in practice this was rare. Statistics are not available nationwide, but Lee, Miguel, and Wolfram ([2020](#)) found that at most 4% of participants in a rural sample in western Kenya randomly selected to receive a free electricity connection chose not to receive one.

Some households preferred to get more than one meter in their compound, for example to leverage the lifeline tariff, or for independence between the households residing in the

⁶ Enumerators only counted unconnected compounds that were within connection distance of the existing electricity network, so this may be an underestimate. [Section 4](#) provides more details on surveying methodology.

compound.

Across all contracting types, 30% of unconnected households noted that they were absent on the day on which Kenya Power enrolled households or when construction happened. 22% of households who did not get connected reported the key barrier as up-front costs: 16% name internal wiring costs and 9% of connected households report having been asked to pay a bribe. This is noteworthy because, according to LMCP media information campaigns during this period, there was not supposed to be any up-front cost (Kenya Power, 2016b): ready-boards were supposed to have been made available to households who were unable to pay the upfront wiring costs, and of course bribes are illegal. While households were surveyed after connections were completed and the analysis is thus unable to strictly separate causation from selection, certain plausibly exogenous household characteristics predict whether a household was connected, including education levels (positively), household size (positively), and distance to transformer (negatively).

C.6 Experiences with bribery

Households also report numerous instances of bribery. In our household survey data, 8% of households connected under LMCP had been explicitly asked for money by the contractor, with amounts generally ranging from \$5 to \$50. Tragically, a small number of households report having paid an individual claiming to be a contractor, only to never hear from them again and to remain unconnected. 5% of unconnected households report not wanting a connection, for example because they are simply not interested in having electricity or because they think electricity is unsafe (this is similar to the rate reported in Lee, Miguel, and Wolfram (2020) noted above).

C.7 Contracting procedure harmonization and similarity

To curtail corruption, IFIs often mandate extensive procurement regulations when financing public investment. WB and AfDB regulations include harmonized master procurement documents and standard bidding documents (WB, 2014; AfDB, 2014):

“Model documents have been developed for bidding of goods, works (small and large contracts), and plants (design, build, install), for requesting consulting services proposals, and for prequalification of bidders.

“Country surveys found... a high level of harmonization among policies of the [WB] and those of the regional development banks and other major donors, including the [AfDB].

“Not only are the policies harmonized (as set out in the Guidelines), but those procedures reflected in standardized bidding documents are also largely harmonized.”
— WB (2014)

Much of these efforts focused on the harmonization of auction procedures and evaluation rules, including the circumstances under which competitive bidding would be required for procurement, which firms would be eligible to bid, and how bids would be evaluated.

For the LMCP, contracts funded by the WB and AfDB largely used the same standard reverse auction format, where the lowest bid wins the contract. Both funders also employed the same approach for updating bids ex post if reasonable unforeseen changes were required, for example, to correct arithmetic or currency conversion errors or to make adjustments to

meet published standards for the goods or services. In line with the procurement regulations that were in effect, an exception was made for contracts of smaller value, for instance for consultancy services (such as the unbundled design contracts), which were selected via quality and cost based scoring. Both funders also used the same set of technical household specifications. Visual inspection of materials and specifications showed that these were almost indistinguishable (see for example the ready boards shown in [Figure A1](#)).

Our research also included dozens of qualitative interviews with staff at Kenya Power, the WB, and the AfDB, as well as independent construction contractors ([Subsection C.1](#) provides a list of individuals consulted). The goal of these conversations was to identify the similarities and differences between contracting procedures across the two donors. The information provided across all these conversations was that the unbundling of contracts and presence of different monitoring activities were far and away the biggest difference between the donors, and that all other aspects of contracting were very similar as a result of the streamlining process described above. The key concern consistently raised by staff at Kenya Power was that—due to the person-hours that go into preparing a bid solicitation document, reviewing bids, and awarding contracts—the large number of WB-funded contracts would require significantly more work.

C.8 Contractors

Contractors that bid on LMCP contracts are generally medium-to-large construction firms with a track record of completed projects. Contractors that won the AfDB- and WB-funded LMCP contracts were a mix of Kenyan firms and international firms, with some joint ventures comprised of two or more firms. To qualify, bidders must satisfy certain requirements related to financial capacity, prior experience including with similarly sized jobs, and any record of sanctioning and litigation.

For the 34 larger-value LMCP contracts including turn-key contracts and those covering materials and installation, Kenya Power used single envelope international competitive bidding, awarding contracts to the lowest-cost bidder. The remaining 13 contracts for consultants (including designers under the unbundled approach) were selected using quality and cost-based selection, a method permitted under both funders' regulations for lower-value consultant services. In the end, Kenya Power awarded the 47 contracts to 41 unique contractors (six contractors won two contracts). Other than a single metering contractor there was no overlap between bundled and unbundled contractors.

WB-funded contracts awarded under competitive bidding (23 non-consultant contracts for materials and installation) attracted an average of 15.6 bids per contract. Among these, installation contracts attracted an average of 7 bids per contract, while materials contracts tended to attract a higher number of bidders. In comparison, AfDB-funded contracts awarded under competitive bidding (11 non-consultant turn-key and meter contracts) attracted an average of 10 bids per contract.

Twenty-one companies were directly awarded at least one unbundled supplier contract and 29 companies were listed as a potential subcontractor for a bundled contractor. In addition to the winning bids, 185 competing bids for unbundled supplier contracts were considered eligible. We focus on poles, cables, and conductors, which were procured through competitive auctions for unbundled contracts but subcontracted out by bundled contractors.

The winners of 11 AfDB-funded turn-key and meter supply contracts had been selected

from 110 bidders. Six of the 10 turn-key contracts winners were Kenyan while four were foreign (Capital Business 2015). The set of contractors awarded WB contracts also included a mix of Kenyan and International firms, with Kenyan firms primarily awarded bids for the supply of wooden and concrete poles. Reflecting the similarities in procurement procedures, the contractors winning WB-funded contracts and the subcontractors being approved for AfDB-funded contracts are similar in terms of country of origin, with around two-thirds being Kenyan, 10% Indian, and 10% Chinese, and the remainder from other countries (Figure A2).

There is no blanket provision preventing firms from submitting—or being awarded—bids with both donors simultaneously. Indeed, many of the AfDB contractors named above have in the past bid on—and in many cases been awarded—WB contracts. International procurement can be thought of as a repeated game: poor contract performance can have serious ramifications on long-term outcomes. Several LMCP contractors have been debarred at least once by the WB or the AfDB (Kenya Power, 2018; Spotlight East Africa, 2020). For example, in October 2018 the WB Sanctions Board imposed “a sanction of debarment” on the Indian company Angelique International for “fraudulent practices as defined in Paragraph 1.16(a)(ii) of the January 2011 Procurement Guidelines.” (WB 2017; WB 2011).

Many of the pole supply firms had existing relations with Kenya Power even prior to the start of the LMCP. As an example, public minutes from a pre-bid meeting for wooden pole procurement organized by Kenya Power in 2014 indicate that eight of the wooden pole suppliers that won WB contracts or AfDB sub-contracts for the LMCP in 2016-2017 were already engaging with Kenya Power as early as 2014, well before the launch of the LMCP (Kenya Power, 2016c), and in many cases even before that (Business Daily, 2007).

Contracts of different sizes (e.g., smaller versus larger) and types (e.g., consultancy, goods, or works) had somewhat different requirements. It is standard practice across funders to require more stringent qualification criteria for contracts of larger value. As a result, qualification criteria for bidders—such as what constitutes evidence of sufficient financial capability—differed across contracts. Domestic suppliers of goods were also permitted a margin of preference in the evaluation of bids, in line with both funders’ procurement regulations. Since only unbundled contracts (those funded by the WB) included contracts specifically for major materials, only the evaluation of bids for poles and cables contracts included a margin of preference for domestic suppliers.

C.9 Oversight

The materials inspections for both funders required detailed mechanical and chemical inspections of 10 poles out of each batch of 500 poles. These visits would usually take place at the physical factory (often located in India, China, or Kenya). However, a number of factory assessments between 2020-2022 had to be conducted via Zoom for public health reasons.

The funders’ oversight structures were similar: the WB’s project manager managed 22 cluster and site supervisors across six offices nationwide, while the AfDB’s project manager managed 19 cluster and site supervisors across four offices nationwide. The consultants’ primary activities during the construction process included conducting site-level spot checks, collecting monthly progress reports from contractors, and hosting (at least) monthly meetings with Kenya Power and each respective contractor.

C.10 Cost-benefit calculations

The cost-benefit calculations in [Section 7](#) make several simplifying assumptions. They value quality differences according to discounted future costs to replace poles at the end of their useful life. The calculations assume that other maintenance costs are similar, despite differences in construction quality. Each pole is assumed to have a constant probability of failure in any given year. The total number of new connections nationwide is assumed to be as reported in [citeKenyaPower20171108](#). Meanwhile, consistent with survey data from the five counties study area, the total number of poles is assumed to be 1.51 times the total number of new connections. We assume a uniform replacement cost of \$100 per pole (for materials alone), consistent with contract amounts and discussion in (Muthike and Ali, [2021](#)). While the procurement cost per pole was different for AfDB and WB contracts during the LMCP, Kenya Power, not the multilateral donor, is responsible for long-term maintenance and repair and would thus procure these items independently. We assume that about half of total replacement costs is for materials alone, which is roughly consistent with contract amounts in the WB Phase I construction.

Bid evaluation and adjustment

[2.49-2.50^a] The purpose of bid evaluation is to determine the cost to the Borrower of each bid in a manner that permits a comparison on the basis of their evaluated cost. Subject to paragraph 2.58, the bid with the lowest evaluated cost, but not necessarily the lowest submitted price, shall be selected for award. The bid price read out at the bid opening shall be adjusted to correct any arithmetical errors. Also, for the purpose of evaluation, adjustments shall be made for any quantifiable non-material deviations or reservations. Price adjustment provisions applying to the period of implementation of the contract shall not be taken into account in the evaluation.

[2.49-2.50^b] The purpose of bid evaluation is to determine the cost to the Borrower of each bid in a manner that permits a comparison on the basis of their evaluated cost. Subject to paragraph 2.58, the bid with the lowest evaluated cost, but not necessarily the lowest submitted price, shall be selected for award. The bid price read out at the bid opening shall be adjusted to correct any arithmetical errors. Also, for the purpose of evaluation, adjustments shall be made for any quantifiable nonmaterial deviations or reservations. Price adjustment provisions applying to the period of implementation of the contract shall not be taken into account in the evaluation.

Factors of bid evaluation

[2.52^a] Bidding documents shall also specify the relevant factors in addition to price to be considered in bid evaluation and the manner in which they will be applied for the purpose of determining the lowest evaluated bid. For goods and equipment, other factors may be taken into consideration including, among others, payment schedule, delivery time, operating costs, efficiency and compatibility of the equipment, availability of service and spare parts, and related training, safety, and environmental benefits. The factors other than price to be used for determining the lowest evaluated bid shall be, to the extent practicable, expressed in monetary terms in the evaluation provisions in the bidding documents.

[2.52^b] Bidding documents shall also specify the relevant factors in addition to price to be considered in bid evaluation and the manner in which they will be applied for the purpose of determining the lowest evaluated bid. For goods and equipment, other factors may be taken into consideration including, among others, payment schedule, delivery time, operating costs, efficiency and compatibility of the equipment, availability of service and spare parts, and related training, safety, and environmental benefits. The factors other than price to be used for determining the lowest evaluated bid shall, to the extent practicable, be expressed in monetary terms, or given a relative weight in the evaluation provisions in the bidding documents.

Payment schedule

[7^e] In respect of plant and equipment supplied from abroad, the following payments shall be made: Twenty percent (20%) of the total CIP amount as an advance payment against receipt of invoice and an irrevocable advance payment security for the equivalent amount made out in favor of the Employer. The advance payment security may be reduced in proportion to the value of the plant and equipment delivered to the site, as evidenced by shipping and delivery documents. Seventy percent (70%) through confirmed letter of Credit of the total or pro rata CIP amount upon Incoterm "CIP", upon delivery to the can-ier within forty-five (45) days after receipt of documents. Five percent (5%) of the total or pro rata CIP amount upon issue of the Completion Certificate, within forty-five (45) days after receipt of invoice. Five percent (5%) of the total or pro rata CIP amount upon issue of the Operational Acceptance Certificate, within forty-five (45) days

[7^f] In respect of plant and equipment supplied from abroad, the following payments shall be made: Twenty percent (20%) of the total CP amount as an advance payment against receipt of invoice and an irrevocable advance payment security for the equivalent amount made out in favor of the Employer. The advance payment security may be reduced in proportion to the value of the plant and equipment delivered to the site, as evidenced by shipping and delivery documents. Seventy percent (70%) of the total or pro rata CIP amount upon Incoterm "CIP", upon delivery to the carrier within sixty (60) days after receipt of documents [Employer to state the required documents, such as a negotiable bill of lading, a non-negotiable sea way bill, an airway bill, a railway consignment note, a road consignment note, insurance certificate, etc]. Five percent (5%) of the total or pro rata amount upon issue of the Completion Certificate, within forty-five (45) days after receipt

Table C2: Representative examples of text similarity scores

(1) World Bank	(2) African Development Bank	(3)	(4)
Any additional information clarification correction of errors or modifications of bidding documents shall be sent to each recipient of the original bidding documents in sufficient time before the deadline for receipt of bids to enable bidders to take appropriate actions	All modifications of bidding documents information clarifications and corrections of errors shall be sent to each recipient of the original bidding documents and all bidders on record in sufficient time before the deadline for receipt of bids to enable bidders to take appropriate actions	0.71	0.93
The bidder shall not be permitted to correct or withdraw material deviations or reservations once bids have been opened	The bidder shall neither be permitted nor invited by the Borrower to correct or withdraw material deviations or reservations once bids have been opened	0.74	0.93
The rights and obligations of the Borrower and the providers of goods and works for the project are governed by the bidding documents and by the contracts signed by the Borrower with the providers of goods and works and not by these Rules or the Financing Agreements	The rights and obligations of the Borrower and the providers of goods works and nonconsulting services for the project are governed by the bidding documents and by the contracts signed by the Borrower with the providers of goods works and nonconsulting services and not by these Guidelines or the Loan Agreements	0.79	0.9
To be acceptable for use in Bankfinanced procurement these procedures shall be reviewed and modified as necessary to assure economy efficiency the provisions included in Section I of these Rules	To be acceptable for use in Bankfinanced procurement these procedures shall be reviewed and modified as necessary to assure economy efficiency transparency and broad consistency with the provisions included in Section I of these Guidelines	0.81	0.92
Where preferences for domestic or regional manufactured goods or for contractors is allowed the methods and stages set forth in Appendix 2 to these Rules shall be followed in the evaluation and comparison of bids	Where preference for domestically manufactured goods or for domestic contractors is allowed the methods and stages set forth in Appendix 2 to these Guidelines shall be followed in the evaluation and comparison of bids	0.82	0.93
Section II of these Rules describes the procedures for ICB	Section II of these Guidelines describes the procedures for ICB	0.82	0.9
In all respects other than advertisement and preferences the ICB procedures shall apply publication of the award of contract as indicated in Paragraph	In all respects other than advertisement and preferences ICB procedures shall apply including the publication of the award of contract as indicated in paragraph 7 of Appendix 1	0.83	0.95
The factors other than price to be used for determining the lowest evaluated bid shall to the extent practicable be expressed in monetary terms or given a relative weight in the evaluation provisions in the bidding documents	The factors other than price to be used for determining the lowest evaluated bid shall be to the extent practicable expressed in monetary terms in the evaluation provisions in the bidding documents	0.83	0.97
This also applies in cases where specialized agencies act as procurement agents	This also applies in cases where UN agencies act as Procurement Agents	0.85	0.91
The period allowed for submission of bids may be reduced to four weeks	The period allowed for submission of bids may be reduced to 4 weeks	0.86	0.98
The Bank may in addition exercise other remedies provided for under the Financing Agreement	The Bank may in addition exercise other remedies provided for under the Loan Agreement	0.86	0.95
The said entrepreneur selected in this manner shall then be free to procure the goods works and services required for the facility from eligible sources using its own procedures	The said concessionaire or entrepreneur selected in this manner shall then be free to procure the goods works and consulting and nonconsulting services required for the facility from eligible sources using	0.87	0.9

D Additional analyses

D.1 Balance tests

D.1.1 Site-level balance tests

[Table D1](#) tests for site-level balance across the study sites using three independent datasets. Columns (1) and (2) show no difference in road distance to the nearest town by financing source among study sites. It also does not persist using the national sample suggesting this is unlikely to reflect intentional selection. ([Table B3](#) shows a 1.6% difference in travel time when considering all sites nationwide, but this difference is sufficiently small that it is unlikely to explain our results.) There does not seem to be a consistent pattern in funder assignment by the transformer’s nearby public facilities ([Table B4](#)).

Table D1: Balance of World Bank- and African Development Bank-funded sites

	Road Distance		VIIRS Radiance	Land Gradient
	(1)	(2)	(3)	(4)
World Bank (=1)	1.17 (1.67)	0.01 (1.06)	0.02 (0.06)	0.57** (0.24)
Observations	366	366	19085	347
Month FE	No	No	Yes	No
Constituency FE	Yes	Yes	Yes	Yes
AfDB Mean	57.97	72.08	.24	4.36
Outcome variable	Minutes	KM		

Notes: Columns (1) and (2) estimate distance in driving minutes and in kilometers from each study site to the nearest ‘major town’ (WRI, 2007; HERE, 2022). Column (3) estimates monthly average site-level nighttime radiance measured using the Visible Infrared Imaging Radiometer Suite (VIIRS) averaged across the 600 meter radius (Elvidge et al., 2017). SEs clustered by site ([Figure A10](#) and [Figure A9](#) show time series). Column (4) estimates average site-level land gradient recorded using the 90-meter Shuttle Radar Topography Mission Global Digital Elevation Model. This table only includes observations from study sites ([Table B3](#) includes all phase 1 sites). Month and constituency FE included where indicated. [Table B3](#) shows the same for all LMCP sites nationwide. * ≤ 0.10 , ** $\leq .05$, *** $\leq .01$.

Levels and trends of pre-LMCP nighttime radiance and levels of land gradient are balanced across sites funded by different donors (see Columns (3) and (4) of [Table B3](#), [Figure A10](#), and [Figure A9](#)). WB-funded sites within our study sample have an average land gradient that is 0.57 degrees higher, relative to the average gradient of 4.4 at AfDB-funded sites ([Table D1](#)). However, the difference in construction quality between WB- and AfDB-funded sites is largely constant across the entire support of land gradient ([Figure A16](#)). Given the observed relationship between construction quality and land gradient, the magnitude of the correlation between land gradient and funder assignment is unable to explain the results

that we find. Still, all site-level regressions include land gradient as a control. [Subsection 6.5](#) discusses the robustness of the results with respect to land gradient in more detail.

D.1.2 Ward-level balance tests

The fraction of sites in a ward that are funded by the WB is not correlated with most pre-LMCP ward-level socioeconomic characteristics, with electoral outcomes, or with ethnic composition ([Table B5](#)). Some of the other coefficients are statistically significant but small in magnitude: for example, shifting from 0% of sites in a ward being WB-funded to 100% of sites in a ward being WB-funded is associated with 4% fewer people with high-quality roofs. A joint F-test cannot reject that the sites are the same along all these characteristics. The correlation between baseline socioeconomic and political characteristics and the assignment of sites to a specific funder appears modest and not related to the outcomes that we study.

Within the five study counties, there is some imbalance in 2009 electrification rates and ward land area ([Table B7](#)). However, the differences are small in magnitude, and when combined with the small magnitude of the relationship between construction outcomes and electrification rates, are unable to explain the magnitude of our results. [Subsection 6.5](#) assesses the robustness of the results to imbalances in 2009 electrification rates in more detail.

D.1.3 Machine Learning assignment predictions

We supplement the standard site-level and ward-level balance tests with Least Absolute Shrinkage and Selection Operator (LASSO) and Decision Tree tests (Mullainathan and Spiess, 2017; Varian, 2014) to evaluate whether any of the variables we consider can predict assignment to WB or AfDB funding. We include a wide range of both ward and site level variables. Across all approaches, the out-of-sample R^2 is consistently below 0.1, suggesting that a broad set of candidate variables cannot accurately predict assignment ([Table B8](#) and [Table B9](#)). The classification accuracy rate of sites to funders is on average 63%: this is only slightly higher than 57%, which is what would result from simply assigning all sites to AfDB, since around 57% of sites in this sample are constructed through AfDB-funded contracts. The most predictive coefficients are baseline electricity access (in the LASSO method) and Kenyatta vote share in the 2013 presidential election (in both methods). However, the balance test discussed above detects no statistically significant differences along these dimensions ([Table B5](#)).

D.1.4 Qualitative descriptions

From June 2016 through July 2022, members of the research team met extensively with dozens of key Kenya Power personnel. This included the General Manager for Connectivity, who was responsible for all of Kenya Power’s new electricity connections, his key managing staff; two Project Managers who oversaw the nationwide LMCP; and senior managers responsible for the WB and AfDB contracting ([Subsection C.1](#) describes a full list of qualitative interviews conducted). For both the WB and AfDB components, Kenya Power employed one full-time staff member to manage the contracting components and a handful of employees to oversee implementation. The staff who held these positions were all certified electrical engineers with similar skill and education levels, all holding at least a bachelor’s degree in electrical engineering. We also read direct correspondence between Kenya Power and dozens

of members of parliament, consisting of hundreds of letters deciding which transformers each phase of the LMCP would include.

Throughout these activities, our explicit goal was to identify a methodology, reason, or process that Kenya Power had used to assign specific sites to be covered by either the WB contracts or the AfDB sites. We were consistently informed that the assignment process among the donors was ad hoc and did not follow any particular allocation rule. Given the overall mandate (to connect all households within 600 meters of a transformer), Kenya Power and the GoK did not appear to see any obvious strategic benefit in having a particular transformer funded by one donor or the other.

D.2 Jaccard and Embedding similarities

To more rigorously evaluate similarities across the documentation, we use two standard text comparison measures: Jaccard similarity and Embedding similarity. Jaccard similarity is a purely lexical comparison of words that makes no attempt to interpret meaning and we therefore interpret it as a more conservative measure of similarity. It is calculated by dividing the number of unique words shared by both texts by the total number of unique words appearing in either text. To allow for differences in exact word choice, we use OpenAI’s embedding model `text-embedding-3-large` to calculate Embedding scores. This captures similarities in semantic content outside of variations in wording or syntax. As examples, the sentences “Any additional information clarification correction of errors or modifications of bidding documents shall be sent to each recipient of the original bidding documents in sufficient time before the deadline for receipt of bids to enable bidders to take appropriate actions” (from AfDB, 2012) and “All modifications of bidding documents information clarifications and corrections of errors shall be sent to each recipient of the original bidding documents and all bidders on record in sufficient time before the deadline for receipt of bids to enable bidders to take appropriate actions” (from WB, 2014) have a Jaccard similarity of 0.71 and an Embedding similarity of 0.93. For reference, Table C2 (shown below) provides some additional examples of sentences with their corresponding Jaccard and Embedding scores.

D.2.1 Jaccard scores

Jaccard similarity captures overlap in word usage without regard to semantic meaning. Specifically, the Jaccard score is the ratio of unique words that appear in both texts to unique words that appear in at least one text. Mathematically, the Jaccard score of sentences S_1 and S_2 is defined as:

$$\frac{\text{Words in } S_1 \cap \text{Words in } S_2}{\text{Words in } S_1 \cup \text{Words in } S_2}$$

Consider the following simple example comparing the sentences ‘Data is the new oil of the digital economy’ and ‘Data is a new oil’ (source: Study Machine Learning, 2025):

$$\frac{\{\text{data, is, new, oil}\}}{\{\text{a, data, digital, economy, is, new, of, oil, the}\}} = \frac{4}{9} = 0.44$$

Since the number of sentences in most sections of the WB regulations is higher than

in the AFDB regulations, we compare each AFDB sentence with all WB sentences in the same section and then calculate the percentage of AFDB sentences that have at least one WB sentence with a Jaccard Similarity greater than 0.8. This is the percentage reported in [Table 1](#).

There are several reasons why we chose to use this methodology rather than compute an aggregate Jaccard similarity across the entire text. First, Jaccard similarity only measures exact word overlap—when texts get longer, two completely different articles can be written using nearly the same set of words. The longer the text, the more likely two highly structured documents on the same topic will converge toward a common set of words regardless of the documents’ actual meaning. This is why narrowing the comparison scope (e.g., within sentences) can sometimes be more informative. Second, the two texts contain many sentences without direct counterparts—some are present in WB but not in AfDB, and vice versa. Since the WB text generally has more sentences than the AfDB text, this could dilute the Jaccard similarity. For example, if the AfDB text is a direct excerpt from WB text but the WB text contains many additional sentences, the denominator increases, and so Jaccard similarity may not capture the degree to which the texts are aligned word-for-word. Of course, sentences without direct counterparts could well result in differences in meaning; Embedding similarity described in the next subsection is well-suited to measure such differences.

D.2.2 Embedding similarities

“Embedding” is a natural language processing (NLP) term that refers to mapping text to a vector of floating-point numbers. This process not only measures word overlap but also captures syntactic and semantic properties. This is how ChatGPT, for example, can understand what people say.

We use OpenAI’s embedding model `text-embedding-3-large` to calculate Embedding scores. This method represents the semantic and syntactic properties of each text as a 3,072-dimensional vector of numbers. We then compute the Cosine similarity to measure the distance between the embeddings of the two documents, where 1 indicates identical vectors and 0 indicates no correlation.

Text analysis using Embedding similarities is more meaningful at the paragraph level, as mapping a small number of words into a 3000-dimensional vector results in high sparsity, making the computed similarity less reliable. This is why [Table 1](#) reports the aggregate Embedding similarity for specific sections rather than computing scores at the sentence level, as is the case with the Jaccard scores.

Due to a cap on the maximum text length, it was not possible to compute an Embedding similarity score for all sections combined. OpenAI’s embedding model can process approximately 7,000 words (8,191 tokens) in a single input. If the text is longer, it needs to be summarized or compressed before embedding. Since the WB and AfDB regulation have sections with clear counterparts, we report an aggregate embedding score that is an average of the scores for each section.

D.3 Additional robustness tests

We explore the extent to which the modest imbalances in land gradient and 2009 electrification rates identified in [Subsection 5.1](#) drive our results ([Figure A17](#) and [Figure A16](#)). Sites with bundled and unbundled contracting have an average difference in land gradient

of 0.6, and each unit difference in land gradient is associated with a 0.05 standard deviation decrease on the construction quality index; interpreting the land gradient correlation causally would suggest an average effect of -0.03 standard deviations on construction quality due to differences in average land gradient. Recall that unbundled sites have an average construction quality that is 0.82 standard deviations higher, an order of magnitude larger.

Similarly, going from 0% sites in a ward being WB funded to 100% of sites being WB funded is associated with 4 percentage points higher electrification rate in 2009. However, 2009 electrification rates do not appear to significantly predict construction outcomes or differences between bundled and unbundled sites (if anything, the sign on this correlation is negative, both with and without constituency fixed effects). Taken together, the difference in construction quality across contracting structures does not appear to be driven by differences in land gradient or baseline electrification rates.

We assess potential endogeneity concerns related to the assignment mechanism raised in [Subsection 5.1](#). First, WB-funded sites have a 13% higher average land gradient. It is plausible that hilliness slows construction and that this difference explains the WB delays. We therefore examine whether land gradient may have caused any of the difference in construction delays by funder assignment. Land gradient is uncorrelated with construction delays, both unconditionally and conditional on funder: the WB delays persist in a stable manner when controlling for land gradient ([Table B28](#)). Furthermore, lag between WB and AfDB is approximately constant across the entire land gradient support ([Figure A17](#)). The difference in land gradient is therefore unlikely to explain the results. Second, WB sites are significantly less likely to be located near a secondary school or religious building, and more likely to be located near a market center or no public facility at all ([Table B4](#)). The gap in timing between WB and AfDB sites is not significantly different across facility types ([Table B29](#)), and the gap in timing between WB and AfDB sites persists when controlling for facility type ([Table B28](#)). All results in [Table 3](#) control for facility type, which do not qualitatively affect the results. Evaluated together, these analyses make it unlikely that baseline differences in facility type contribute meaningfully to the results.

The GridWatch devices recorded data between June 2021 and June 2022, even though stringing at most AfDB sites was completed between 2017 and 2019 and stringing at most WB sites was completed between 2018 and 2020. Thus, the GridWatch data measured WB sites when they were on average one year newer than the AfDB sites surveyed at the same time. If the aging of the grid negatively affects reliability and voltage quality, then this bias would favor WB in the results. [Figure D1](#) confirms that voltage quality is constant over time, and that the lack of difference in voltage quality between the WB and the AfDB persists even among sites where the time since stringing completion was approximately equal.

For Outcome 4 measuring household installation quality ([Table B17](#)) we replicate the index omitting the question asking the respondent whether they have a ready board, since it is not obvious whether the presence of a ready board is a positive or negative component. Its presence simultaneously indicates Kenya Power provisions and a lack of household preparedness (see [Section 2](#) for more detail).

Of the 250 sites that we surveyed, 26 are located less than 1,200 meters from another site. Given that AfDB sites saw construction on average earlier than WB sites, this could reduce construction at WB sites, as the subset of that site's unconnected households that lie within the 600 meter radius of the nearby site might already have been connected. This

could explain why Table 2 indicates less construction at WB sites. To test this, we replicate this table excluding the 26 sites—12 AfDB and 14 WB—that are within 1,200 of another site. Table B31 shows the results. If anything, the gap between construction at WB and AfDB sites is even larger.

Finally, the private contractor awarded lots 3 and 5 of the WB construction contracts⁷ experienced unusual financial circumstances and this may have interfered with the timeliness and quality of their construction. We therefore repeat the analysis from Table 3 excluding these contracts, and then only keeping a balanced panel of counties. This does not affect results: if anything, household installation quality and reliability and safety were slightly worse at the remaining WB sites, although the results are noisier (Table B30).

D.4 Conceptual framework

To illustrate how the degree of contract bundling and the extent of monitoring can affect outcomes, we present a simple framework through which to analyze these contracting structures. To fix ideas, assume the principal wants the three components (designs, supplies, and installation) to be carried out by firms selected from a continuum of firms $\gamma_i \in [0, \infty)$ defined by firm type, where higher types produce higher quality at higher cost. Firms are selected through a competitive auction, and only firms exceeding an exogenously set firm type threshold $\bar{\gamma}$ are eligible to bid (in our setting, bidders must meet uniform global WB and AfDB documentation requirements).⁸ Each firm also chooses how much effort $e_i \in [0, \infty)$ to exert. The quality of each component is determined by the firm type and the effort that firm exerts: $q_i = \gamma_i + e_i$. Overall project quality is the sum of the three components: $Q = q_a + q_b + q_c$.

Increasing firm type γ_i incurs convex cost $c(\gamma_i)$. A firm exerting effort e_i incurs convex cost $d(e_i)$, with $d'(0) = 0$ and $d(0) = 0$. For a given level of quality q , denote $\gamma^*(q)$ and $e^*(q)$ to be the cost-minimizing combination of effort and firm type, with $\gamma^* > \bar{\gamma}$. We assume perfect competition, with firms bidding to cover their costs, $b_i = c(\gamma_i) + d(e_i)$ (a reasonable approximation in our context, where most auctions attract a large number of bids from many local and international firms).

The principal faces two problems. First, it has imperfect information about firm types, and can only observe whether a firm exceeds the minimum threshold (installers, on the other hand, have perfect information about designer and supplier types⁹). Second, since firms decide how much effort to exert after contracts have been awarded, the principal must

⁷ A single consortium won both of these contracts.

⁸ An alternative model could endogenize this threshold: lower monitoring or contract unbundling might change the optimal threshold. We omit that here as the threshold is exogenously set by donor policy in this context.

⁹ Conducting installation using another firm’s designs and supplies provides a level of insight into that firm’s type and effort that cannot be achieved by the principal through even very careful monitoring. Furthermore, in practice many installers collaborate with the same designer or supplier much more frequently than the principal does.

find a way to incentivize effort.

The principal has two contracting tools. First, it can choose to offer either one bundled contract ($t = 1$) or three unbundled contracts ($t = 3$). Second, the principal can implement either low ($m = L$) or high ($m = H$) monitoring. Under low monitoring, all contractors are paid regardless of realized quality. In this case, firms exert no effort, and quality simply corresponds to their firm type: $q_i = \gamma_i$. Under high monitoring, contractors are only paid if the quality of the component(s) included in their contract meets a quality threshold $\bar{q}$. If any component is produced below output quality $\bar{q}$, then the contracted firm is not paid. The principal implements the auction as follows, where P and A denote decisions taken by the Principal and Agent (firms), respectively:

- (P1) The principal sets the auction parameters (t and m). For unbundled contracts, the principal runs three sequential auctions. For bundled contracts, the principal runs one auction for installation.
- (A1) Each eligible firm decides whether or not to bid. If it chooses to bid, it then chooses a bid amount b_i .
- (P2) The principal identifies the lowest eligible bid as the winner of each auction.
 - If the contracts are unbundled, the principal awards three contracts.
 - If the contract is bundled, the principal awards one contract for installation. The winner then selects a designer $\gamma_a \in [0, \infty)$ and a supplier $\gamma_b \in [0, \infty)$ with full discretion (they can choose firms below the threshold: $\gamma_i < \bar{\gamma}$). The installer can mandate their effort levels e_a and e_b .
- (A2) Each firm chooses an effort level e_i and realizes their output quality q_i .
- (P3) The principal pays the contractor(s) (or not).
 - If $m = L$, all firms are paid regardless of q_i .
 - If $m = H$ and $t = 1$, all firms are paid only if $q \geq \bar{q}$ for each component (and the installer pays the designer and supplier *iff* it is paid).
 - If $m = H$ and $t = 3$, the designer, supplier, and installer are each paid only if their respective output quality is at least $\bar{q}$.

Assuming perfect competition, firms bid to cover their costs:

- If $t = 1$, the firm's bid is $b_c = c(\gamma_a) + d(e_a) + c(\gamma_b) + d(e_b) + c(\gamma_c) + d(e_c)$. The firm pays the design firm $c(\gamma_a) + d(e_a)$ and the supplies firm $c(\gamma_b) + d(e_b)$.
- If $t = 3$, each firm's bid is $b_i = c(\gamma_i) + d(e_i)$.

Case 1 ($t = 3, m = L$):

- Firms with $\gamma_i = \bar{\gamma}$ will bid $c(\bar{\gamma})$ and win all three auctions.
- There is no incentive to supply any effort, so $e_i = 0$
- Project quality will be $3\bar{\gamma}$, with cost per contract $b(3, L) = c(\bar{\gamma})$ and total cost $3c(\bar{\gamma})$

Case 2 ($t = 3, m = H$):

- For each component, the winning firm is the one that can produce output quality $\bar{q}$ at lowest cost, since no other firm can bid lower than them while achieving at least zero profit. The winning firms will each have type γ_i and choose e_i such that: $c'(\gamma_i) = d'(e_i)$. Let γ^* and e^* be the solution to this problem.

- Project quality will be $\bar{q}$, with cost per contract $b(3, H) = c(\gamma^*) + d(e^*)$ and total cost of $3c(\gamma^*) + 3d(e^*)$.

Case 3 ($t = 1, m = L$):

- The winning firm has $\gamma = \bar{\gamma}$ and bids $c(\bar{\gamma}) + 2c(0)$ in the auction. It is the firm of at least type $\bar{\gamma}$ that can bid the lowest. It then exerts no effort.
- For design and materials, it has no incentive to select firms with more than minimum firm type, and it will contract assuming zero effort from those firms.
- Project quality will be $\bar{\gamma}$, with total cost $b(1, L) = c(\bar{\gamma}) + 2c(0)$.

Case 4 ($t = 1, m = H$):

- With full ability to contract on firm type and effort exerted, the winning firm will wish to supply output quality $\bar{q}$ at minimum cost. The winning firm that is able to bid the lowest will have γ_c^* and choose effort level e_c^* ; and will seek to contract design and materials firms with type and levels of effort $\gamma_a^*, e_a^*, \gamma_b^*, e_b^*$ such that: $c'(\gamma_a) = d'(e_a) = c'(\gamma_b) = d'(e_b) = c'(\gamma_c) = d'(e_c)$.
- Project quality will be $\bar{q}$, with total cost $b(1, H) = 3c(\gamma^*) + 3d(e^*)$.

Contractors often also look to bid on future donor-funded projects as well, and under-performance or fraud can lead to sanctions that make them ineligible for future projects. Therefore one could also conceptualize equilibrium in this setting as a repeated game, where each round is played as described above, and each player pursues a grim trigger strategy.

While unbundling and higher monitoring may improve quality, they incur administrative costs $\kappa(m, t)$, which are increasing in both arguments. Furthermore, monitoring incurs delays both directly (the monitoring itself takes time) and indirectly (incurring greater effort may cause firms to complete activities more slowly), and unbundling causes delays by adding administrative and coordination tasks associated with conducting three sequential sets of auctions rather than a single set of auctions. These delays lower the project's future welfare gains W according to a function $D(m, t, \delta) < 1$, where δ denotes the principal's intertemporal discount rate. When selecting t and m the principal wishes to maximize net benefits, factoring in these costs and delays, contract costs b_i , and long-term maintenance costs M , which decrease with aggregate project quality Q (the sum of the qualities of the three components: $Q = q_a + q_b + q_c$). Maintenance costs can be thought of as the expenses needed to maintain project benefits at level W over the lifetime of the project (from year $y = 1$ to Y); higher quality construction lowers these expenses.

$$\text{Project Net Benefits} = \underbrace{D(m, t, \delta)W}_{\text{Welfare gains}} - \underbrace{\kappa(m, t)}_{\text{Administrative costs}} - \underbrace{\sum_{i=1}^t b_i(m, t)}_{\text{Contract costs}} - \underbrace{\sum_{y=1}^Y \delta^y M(Q(m, t))}_{\text{Long-term maintenance costs}} \quad (2)$$

The principal's optimal choice of monitoring and (un)bundling will depend on, for example, the delays incurred, their discount rate, and the effect of quality on long-term maintenance costs.

D.5 Long-term resilience of the grid

Construction might affect resilience through two key engineering channels. First, voltage quality tends to worsen with distance from the central transformer.¹⁰ We find that this is primarily due to the increasing number of customers connected more closely to the transformer rather than the distance traveled along the LV electricity wire per se. [Table D2](#) shows no difference between funders in distance resilience.

Panel A of [Figure D1](#) explores the correlation between 10th percentage of voltage quality and distance to the transformer along the LV network.¹¹ There does not appear to be a significant or discontinuous decline after 600 meters, the eligibility cutoff for a subsidized LMCP household connection, suggesting greater returns to scale might have been achieved under a higher distance eligibility cutoff.

Second, voltage quality could worsen with the passage of time, as infrastructure ages. Higher quality construction might make infrastructure more resilient and slow any associated decay. The time since construction varies across our sample since stringing was completed between June 2017 and January 2021, while GridWatch devices recorded data between June 2021 and June 2022. Panel B of [Figure D1](#) examines the correlation between voltage quality and time since construction. At both AfDB and WB sites, the grid appears resilient to aging for the first five years after the completion of stringing.

¹⁰ Jacome et al. ([2019](#)) find a similar result in Zanzibar, Tanzania.

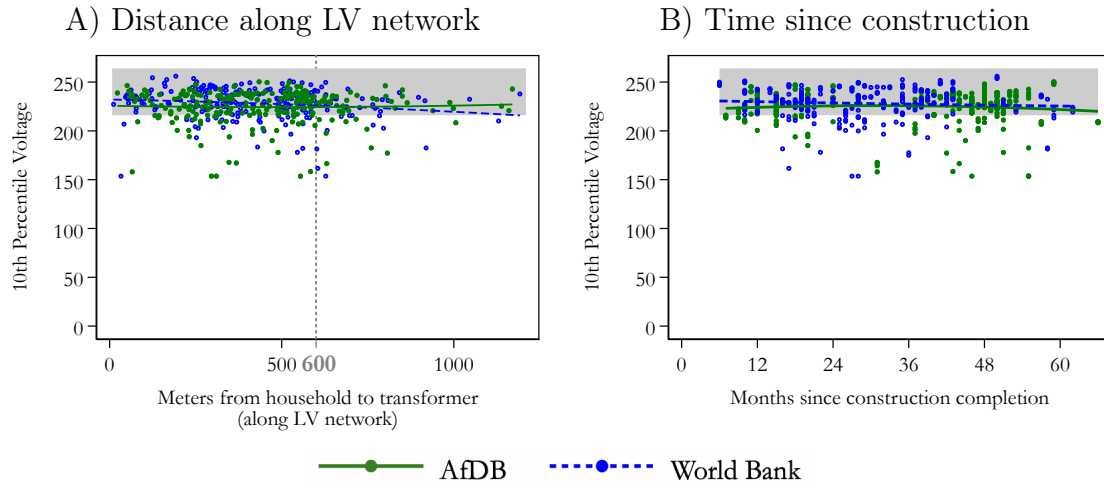
¹¹ The results look similar when using mean voltage. Using the 10th percentage of voltage quality is in line with engineering expectations around how resilience might affect voltage quality.

Table D2: Resilience of voltage to distance from transformer

	(1)	(2)	(3)
Distance Along Wire	-0.000 (0.003)	-0.000 (0.003)	-0.000 (0.003)
Customer Connections	-0.490*** (0.160)	-0.490*** (0.163)	-0.615*** (0.230)
World Bank		0.043 (1.305)	-0.788 (2.741)
World Bank=1 $\times$ Distance Along Wire			-0.002 (0.008)
World Bank=1 $\times$ Customer Connections			0.261 (0.347)
Constant	237.937*** (1.345)	237.918*** (1.459)	238.452*** (1.507)
Observations	377314	377314	377314
Control Mean	235.69	235.69	235.69

Notes: Standard errors are clustered by respondent and shown in parentheses. * ≤ 0.10 , ** $\leq .05$, *** $\leq .01$.

Figure D1: Voltage quality resilience to distance and infrastructure aging



Notes: 10th percentile of hourly voltage readings with quadratic fit line. The gray area indicates Kenya's nominal voltage, 240 V, $\pm 10\%$ as per international utility guidelines. Panel A explores how a household's distance to the central transformer (as measured along the LV network) affects voltage quality. Panel B explores how the passage of time since the initial completion of construction affects voltage quality. Neither appear to strongly affect voltage quality. WB and AfDB exhibit similar trends.

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